

EvoLen: Evolution-Guided Tokenization for DNA Language Model

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Abstract

Tokens serve as the basic units of representation in DNA language models (DNALMs), yet their design remains under-explored. Unlike natural language, DNA lacks inherent token boundaries or predefined compositional rules, making tokenization a fundamental modeling decision rather than a naturally specified one. While existing approaches like byte-pair encoding (BPE) excel at capturing token structures that reflect human-generated linguistic regularities, DNA is organized by biological function and evolutionary constraint rather than linguistic convention. We argue that DNA tokenization should prioritize functional sequence patterns like regulatory motifs—short, recurring segments under evolutionary constraint and typically preserved across species. We incorporate evolutionary information directly into the tokenization process through **EvoLen**, a tokenizer that combines **E**volutionary stratification with **L**ength-aware decoding to better preserve motif-scale functional sequence units. EvoLen uses cross-species evolutionary signals to group DNA sequences, trains separate BPE tokenizers on each group, merges the resulting vocabularies via a rule prioritizing preserved patterns, and applies length-aware decoding with dynamic programming. Through controlled experiments, EvoLen improves the preservation of functional sequence patterns, differentiation across genomic contexts, and alignment with evolutionary constraint, while matching or outperforming standard BPE across diverse DNALM benchmarks. These results demonstrate that tokenization introduces a critical inductive bias and that incorporating evolutionary information yields more biologically meaningful and interpretable sequence representations. Code, pretrained and fine-tuned checkpoints, and tokenizer files are available at <https://github.com/HN020719/EvoLen> and <https://huggingface.co/EvoLenTokenizer>.

1 Introduction

DNA language models (DNALMs) require transforming continuous nucleotide sequences into discrete tokens. Unlike natural language, DNA does not come with intrinsic boundaries analogous to words, so tokenization is not a neutral preprocessing step. Instead, it determines the units through which sequence structure is exposed to the model, and therefore shapes what patterns can be represented efficiently. Most existing genomic tokenization strategies inherit NLP paradigms: fixed-length k -mers (Ji et al., 2021) impose uniform scales, while adaptive methods like byte-pair encoding (BPE) (Sennrich et al., 2016), applied to DNA (Zhou et al., 2023), optimize for frequency and compression. However, these objectives often ignore biological function, leading to the fragmentation of coherent functional units (Provilkov et al., 2019).

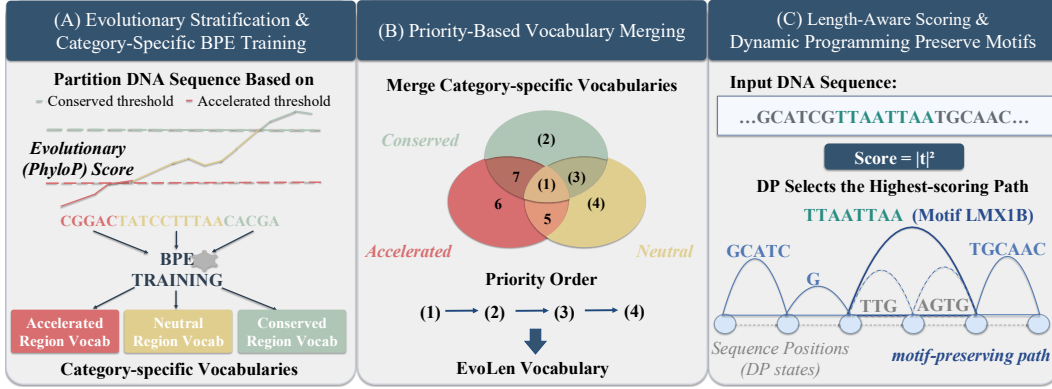


Figure 1: **EvoLen tokenization pipeline.** **A)** Partition the genome into conserved, neutral, and accelerated regions using phyloP scores and train a category-specific BPE tokenizer on each sequence pool. **B)** Merge the resulting vocabularies using a conservation-prioritized rule. **C)** Apply length-aware scoring to the merged vocabulary and dynamic programming to ensure optimal, non-overlapping segmentation. This enables EvoLen to preserve more functional motifs as intact tokens than BPE, enhancing representation coherence.

This mismatch is especially important in regulatory genomics, where many predictive signals are carried by short functional sequence patterns. A key example is transcription factor (TF) binding motifs—short DNA segments (typically 6–12 bp) recognized by regulatory proteins that control gene expression (Avsec et al., 2021). When BPE splits such a pattern into multiple pieces or merges part of it with flanking sequence, the intact functional unit is no longer explicitly represented, making the full regulatory pattern harder for the model to learn. As illustrated by the examples in Appendix Figure 5, standard BPE breaks the LMX1B binding sequence TAAATTAA into shorter substrings and merges part of the motif with neighboring bases. These limitations suggest that genomic tokenization should account for biological structure rather than substring frequency alone.

One broadly available signal for such structure is evolutionary constraint: functional genomic elements are more likely to be preserved across species, whereas less constrained or lineage-specific regions may evolve more freely. Several signals could in principle capture this structure, including chromatin accessibility, TF binding annotations, or comparative genomics-based conservation scores such as phyloP (Pollard et al., 2010; UCSC Genome Browser, 2024b), phastCons (Siepel et al., 2005), and GERP (Davydov et al., 2010). Among these, phyloP is particularly well suited to tokenizer construction because it is available at single-nucleotide resolution, provides signed scores that distinguish conserved from accelerated positions, and is derived from cross-species sequence comparison rather than from any downstream task label. This makes it a genome-wide, fine-grained, and task-agnostic signal for identifying sequence regions likely to carry functional information.

This motivates a central question: *what properties should tokens satisfy to better capture regulatory sequence structure?* To address this, we formalize these desired properties in Section 2 and introduce **EvoLen**, a tokenizer that integrates **E**volutionary stratification with **L**ength-aware decoding through a **partition-and-merge** strategy. As illustrated in Figure 1, EvoLen first **partitions** the genome into three sequence pools based on phyloP scores—representing highly conserved regions, where substitutions are rare across species and which typically carry function; weakly constrained (neutral) regions, which evolve at the background rate; and rapidly evolving (accelerated) regions, which change faster than the background. It then trains separate BPE tokenizers on each pool and **merges** the resulting vocabularies into a single set using a priority rule that favors tokens from evolutionarily preserved regions. Finally, EvoLen applies length-aware scoring to the merged vocabulary and dynamic programming to ensure optimal, non-overlapping sequence segmentation.

Both token structural evaluations and downstream benchmarks confirm that **EvoLen** preserves functional sequence units while maintaining a flexible subword vocabulary, providing a critical inductive bias for genomic modeling. Our main contributions are as follows:

1. **EvoLen Tokenizer:** We introduce a conservation-aware tokenizer that incorporates evolutionary information into vocabulary construction via a **partition-and-merge** strategy and utilizes length-aware decoding for optimal segmentation.
2. **Biological Interpretability:** We demonstrate that **EvoLen** yields more biologically meaningful representations. At a vocabulary size of 5,120, it preserves transcription factor (TF) motifs as single tokens 27.4% more frequently than standard BPE. Furthermore, it enhances regulatory specificity, increasing the Jensen–Shannon distance between promoter and enhancer token-length distributions by 47% , and improves alignment with evolutionary signals by up to 14.4%.
3. **Downstream Performance:** We show that informed tokenization alone improves genomic modeling without altering model architecture or pre-training objectives. Despite using a smaller pretraining budget, **EvoLen** outperforms the baseline on 11 of 15 task groups and ranks in the top two among four tokenization strategies on 13 groups, including TF binding and cross-species brain cell-type prediction.

2 Problem Setup

Given a DNA sequence $S = (s_1, s_2, \dots, s_n)$, a tokenizer maps S to a sequence of contiguous tokens $(t_1, t_2, \dots, t_k)$. This segmentation defines the basic units available to the model and therefore determines the scale at which sequence patterns can be represented and composed. For genomic sequence modeling, our goal is to construct tokens that better align with biological features; the four structural properties below define the criteria that guide the EvoLen pipeline in Figure 1 and the evaluations in Section 4. In particular, desirable genomic tokens should satisfy: **(P1) Functional Integrity:** preserve short functional units such as transcription factor binding motifs; **(P2) Regulatory Specificity:** produce tokenization patterns that differ across genomic contexts such as promoters and enhancers; **(P3) Evolutionary Consistency:** align token boundaries with regions under similar evolutionary constraint; **(P4) Pattern Recurrence:** capture recurrent functional sequence patterns rather than merely frequent substrings. Our goal is therefore to construct a tokenizer whose vocabulary construction and decoding procedure better satisfy these properties than standard frequency-driven tokenization.

3 EvoLen Tokenizer Construction

We now describe how EvoLen is constructed. The tokenizer consists of three components corresponding to the pipeline in Figure 1: evolutionary stratification with category-specific BPE training, conservation-prioritized vocabulary merging, and length-aware decoding via dynamic programming.

3.1 Evolutionary Stratification and Category-Specific BPE

Why stratify the genome? Different parts of the genome are shaped by different evolutionary pressures. Conserved regions are more likely to contain functionally constrained elements, whereas neutral and accelerated regions reflect different sequence dynamics. If all genomic regions are pooled together during token discovery, these signals can be blurred, making it harder for the tokenizer to recover biologically coherent units.

How we stratify. We therefore use phyloP scores (UCSC Genome Browser, 2024b), which quantify deviations in substitution rate across species, to partition the human genome (UCSC Genome Browser, 2024a) into three evolutionary categories: conserved (con), neutral (neu), and accelerated (acc). To obtain stable regional signals, we first divide the genome into non-overlapping 100 bp bins and compute the mean phyloP score within each bin. Let

x_b denote the mean phyloP score of bin b . We then compute the global mean μ and standard deviation σ over all bins, and assign each bin using a two-tailed Z-score rule:

conserved: $x_b > \mu + z\sigma$, accelerated: $x_b < \mu - z\sigma$, neutral: $\mu - z\sigma \leq x_b \leq \mu + z\sigma$,

where $z = 1.645$, corresponding to a two-tailed significance level of $p < 0.1$. Thus, bins with significantly positive phyloP values are assigned to the conserved category, bins with significantly negative values are assigned to the accelerated category, and the remaining bins are treated as neutral.

This procedure yields three sequence pools with distinct evolutionary profiles. Conserved bins are enriched for regions under purifying selection, accelerated bins capture regions with elevated substitution rates, and neutral bins provide a broad genomic background.

Category-specific BPE training. We then independently train a BPE tokenizer on each of the three sequence pools to learn a candidate vocabulary for that evolutionary regime. This produces three category-specific vocabularies that capture distinct merge patterns, rather than forcing a single tokenizer to absorb all sequence contexts at once.

3.2 Priority-Based Vocabulary Merging

Why build vocabularies separately first? After stratification, each evolutionary category contains a different mix of sequence patterns. Conserved regions are more likely to contain reusable functional units, whereas neutral and accelerated regions contribute broader background diversity. Training a single tokenizer across all regions can blur these differences. We therefore first learn candidate vocabularies within each category and then merge them using a conservation-prioritized rule that favors reusable sequence structure.

How we merge them. For each category $c \in \{\text{con}, \text{neu}, \text{acc}\}$, we learn a candidate vocabulary V_c . We then construct the final vocabulary using the priority order illustrated in Figure 1B: (1) tokens shared across all three regions; (2) conserved-specific tokens; (3) tokens shared by conserved and neutral regions but not accelerated regions; and (4) neutral-specific tokens to fill the remaining capacity.

This rule biases the final vocabulary toward sequence units that are stable across functionally constrained genomic contexts while still retaining sufficient coverage of the broader genome. In practice, it favors tokens that are more likely to reflect biologically reusable structure rather than purely local frequency artifacts.

3.3 Length-Aware Scoring and Dynamic Programming Decoding

Why is an additional decoding step needed? Even after conservation-aware vocabulary construction, segmentation can still over-prefer short, frequent substrings. A biologically coherent motif may already exist in the vocabulary, yet a standard frequency-based decoder can still split it into shorter pieces if those pieces receive better local scores. This creates a mismatch between what enters the vocabulary and what is actually used during tokenization. To reduce this unnecessary fragmentation and better preserve functional integrity, EvoLen adds an explicit length-aware preference during decoding.

Scoring rule. After vocabulary construction, the merged token set is serialized as a Uni-gram tokenizer (Kudo, 2018; Kudo & Richardson, 2018). Each token t is assigned a score proportional to the square of its length, $\text{score}(t) = |t|^2$, where $|t|$ denotes the number of nucleotides in token t . This scoring rule does not force long tokens everywhere. Instead, it rewards longer units when they are already supported by the vocabulary, while still allowing short tokens when no coherent multi-base pattern is available.

Dynamic programming decoding. Given a DNA sequence $s = s_1s_2 \cdots s_n$ and a scored vocabulary V , we use dynamic programming to select the globally optimal non-overlapping segmentation of s . For a sequence prefix ending at position i , let $\text{DP}[i]$ denote the maximum achievable total score for the prefix $s_{1:i}$, with $\text{DP}[0] = 0$. We then compute

$DP[i] = \max_{j < i, s_{j+1:i} \in V} (DP[j] + |s_{j+1:i}|^2)$. This guarantees that the final tokenization is globally optimal under the length-aware objective.

By construction, the decoder prefers longer intact tokens over fragmented alternatives spanning the same span, which in practice reduces fragmentation into short high-frequency substrings and helps preserve motif-scale functional units. This places EvoLen’s decoder in the same family as shortest-path tokenizers: UnigramLM performs Viterbi decoding over a scored vocabulary (Kudo, 2018), and PathPiece makes the shortest-path formulation explicit (Schmidt et al., 2024). EvoLen differs in its scoring function, which rewards length rather than corpus likelihood.

Together, this three-stage pipeline better preserves motif boundaries and yields more coherent sequence representations, as illustrated by the intact motif examples in Appendix Figure 5.

4 Functional and Evolutionary Token Analysis

We next test whether EvoLen satisfies the four properties defined in Section 2. Specifically, we ask whether it better preserves short functional sequence units (P1), produces regulatory element-specific tokenization patterns (P2), aligns with evolutionary constraint (P3), and captures recurrent functional sequence structure (P4). P1, P2 and P4 are evaluated against signals held out from construction, including JASPAR motifs, promoter/enhancer/exon/intron annotations, and a neutral intronic background. None of these are seen by the tokenizer during vocabulary learning. P3 uses phyloP, which does enter construction, but only at the level of 100 bp pools, whereas the evaluation measures per-token conservation, a granularity neither BPE merging nor length-aware decoding optimizes directly.

4.1 Motif Preservation

Why this matters. Many regulatory functions in DNA are mediated by short sequence patterns called TF motifs (Alipanahi et al., 2015; Kelley et al., 2018). A tokenizer that preserves such a motif as one token presents it to the model as a single coherent unit; a tokenizer that splits it forces the model to reconstruct that signal across multiple tokens. We therefore evaluate whether EvoLen better preserves intact motif-scale units, corresponding to **(P1) Functional Integrity**.

How we evaluate it. For each vocabulary size, we compute the **perfect match rate**: the fraction of known motifs that are encoded as a single token without being split. We use transcription factor motifs from the JASPAR 2024 vertebrate library (Bailey et al., 2009; Grant et al., 2011), represented as PWMs, and convert them into fixed motif sequences by thresholding each position at 0.5, trimming wildcard positions at both ends, and retaining the highest-probability nucleotide at the remaining positions. We restrict the analysis to motifs of length at most 12 bp (Zhou et al., 2025). This directly measures how often the tokenizer preserves a complete functional sequence element as one representational unit. Additional motif tokenization diagnostics, including fragmentation, coverage, and consistency across motif variants, are reported in Appendix Figure 4.

Main result. EvoLen achieves a higher perfect match rate than baseline BPE at every vocabulary size, with relative gains of +9.1% at 2,048, +3.8% at 3,072, +17.5% at 4,096, and +27.4% at 5,120 (Figure 2)A. The largest improvement appears at the vocabulary size used for downstream evaluation, where EvoLen preserves substantially more motifs as intact tokens. Overall, these results support **(P1) Functional Integrity**: EvoLen more often represents short functional DNA elements as single units rather than fragmented substrings.

4.2 Functional Region Token-Length Signatures

Why this matters. Different regulatory elements play different roles. Promoters help initiate gene expression (Seizl et al., 2011), enhancers help regulate when and where genes are activated, and exons encode the transcribed sequence itself. If a tokenizer captures the

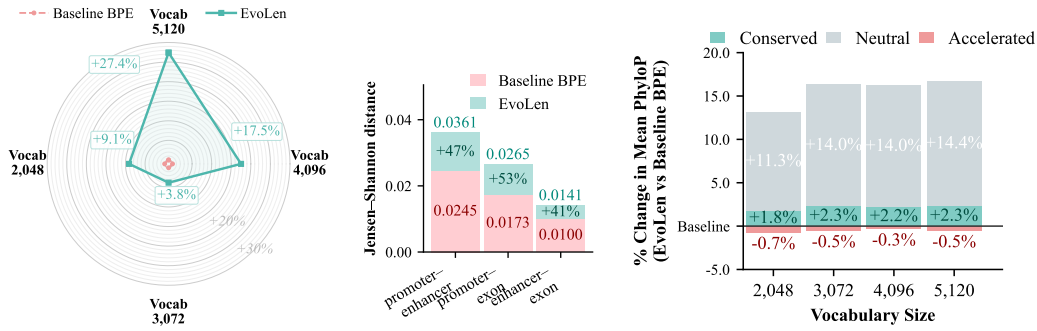


Figure 2: **(A)** EvoLen increases the fraction of motifs preserved as single tokens across all vocabulary sizes (P1). **(B)** Increased Jensen-Shannon distances at vocabulary size 5,120 demonstrate that EvoLen produces more distinct token-length distributions between promoters, enhancers, and exons than BPE (P2). **(C)** Relative gains in mean phyloP scores signify enhanced alignment with evolutionary conservation (P3).

feature of these biological structures, it should not segment all of these regions in the same way. We therefore test whether EvoLen produces more distinct tokenization patterns across various regulatory elements, corresponding to **(P2) Regulatory Specificity**.

How we evaluate it. At vocabulary size 5,120, we compare token-length distributions in promoters, enhancers, and exons. Token lengths are grouped into four bins (1–2 bp, 3–5 bp, 6–8 bp, and 9+ bp), and we measure the Jensen-Shannon distance between each pair of regions (Figure 2)B. Higher distance means the tokenizer produces more distinct length profiles across genomic contexts.

Main result. EvoLen produces higher region-to-region separation than baseline BPE for all three comparisons: promoter–enhancer increases from 0.0245 to 0.0361 (+47%), promoter–exon from 0.0173 to 0.0265 (+53%), and enhancer–exon from 0.0100 to 0.0141 (+41%). This indicates that EvoLen assigns more distinct tokenization patterns to different functional regions. Overall, these results support **(P2) Regulatory Specificity**: EvoLen better distinguishes genomic contexts through its segmentation behavior. Full token-length distributions and pairwise divergences are reported in Appendix A.3.

4.3 Evolutionary Conservation Alignment

Why this matters. Some genomic regions are more evolutionarily constrained than others: conserved regions tend to preserve function across species, while accelerated regions evolve more rapidly (Kircher et al., 2014; Gulko et al., 2015). Because EvoLen uses evolutionary information during tokenizer construction, we expect its tokens to align more closely with these constraint patterns. This directly tests **(P3) Evolutionary Consistency**.

How we evaluate it. We tokenize the same genome with baseline BPE and EvoLen, then ask how evolutionarily coherent the resulting tokens are. For each token, we average the phyloP scores of all bases it spans, and then group tokens by whether they come from conserved, neutral, or accelerated regions (Figure 2)C. If a tokenizer better respects evolutionary structure, its tokens should have higher mean phyloP in conserved regions, more negative mean phyloP in accelerated regions, and clearer separation between categories. We also examine intra-token phyloP variance to test whether any improvement reflects better alignment rather than greater heterogeneity within tokens.

Main result. EvoLen produces tokens whose average conservation scores better match the evolutionary category they come from. In conserved regions, EvoLen tokens have consistently higher mean phyloP than baseline, with gains of +1.8% to +2.3% across vocabulary sizes. In neutral regions, the improvement is larger, at +11–14%, suggesting

that EvoLen better groups weakly constrained sequence patterns that baseline BPE tends to fragment. In accelerated regions, EvoLen yields slightly more negative mean phyloP values than baseline, indicating clearer separation from conserved sequence. Intra-token phyloP variance remains comparable between methods, showing that these gains come from better evolutionary alignment rather than noisier token composition. Together, these results support **(P3) Evolutionary Consistency**: EvoLen produces tokens that better respect the underlying evolutionary structure of the genome. Per-region phyloP statistics across all vocabulary sizes are reported in Appendix A.2.

4.4 Functional Sequence Enrichment

Why this matters. A tokenizer can learn frequent substrings without learning biologically meaningful ones. To test whether EvoLen captures recurrent functional sequence structure instead of generic high-frequency patterns, we ask whether its tokens show clearer enrichment in biologically distinct genomic contexts. This evaluates **(P4) Pattern Recurrence**.

How we evaluate it. We first divide the genome into 12 sequence bins by crossing four genomic regions (promoter, enhancer, exon, intron) with three conservation categories (conserved, neutral, accelerated). For each bin, we tokenize all sequences assigned to that region-conservation combination and compute token frequencies. We then compare each bin against a fixed background consisting of **neutral and intronic sequences**, which serve as a broad baseline with relatively weak functional and evolutionary constraint (Figure 3). We summarize this comparison using the mean $\log_2$ fold-change of token frequencies relative to that neutral intronic background. Less negative values indicate relative enrichment in a given context, whereas more negative values indicate stronger depletion. A biologically meaningful tokenizer should show clearer separation between conserved functional regions and accelerated sequence. Enrichment computation details and diagnostics are provided in Appendix A.4.

Main result. EvoLen shows stronger context-specific enrichment than baseline. In conserved regions, depletion becomes substantially weaker for promoters (from -0.84 to -0.53), enhancers (from -0.58 to -0.14), and exons (from -0.65 to -0.23), indicating that EvoLen better captures the sequence patterns characteristic of functional elements under constraint. EvoLen also increases the separation between conserved and accelerated sequence in most regions, where we define separation as $\Delta_{\text{sep}} = |\bar{E}_{\text{conserved}} - \bar{E}_{\text{accelerated}}|$. Enhancer separation rises from 0.08 to 0.62 , exon separation from 0.15 to 0.35 , and intron separation from 0.29 to 0.42 , while promoter separation remains comparable (0.20 vs. 0.19). Overall, these results support **(P4) Pattern Recurrence**: EvoLen captures recurrent functional sequence structure more clearly than a purely frequency-driven tokenizer.

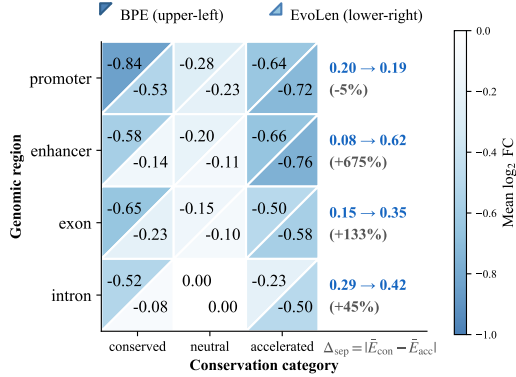


Figure 3: Token enrichment (mean $\log_2$ fold-change) relative to neutral intronic background, crossed by genomic region and conservation category (P4).

5 Downstream Evaluation

Section 4 showed that EvoLen better preserves functional sequence units, aligns more closely with evolutionary constraint, and produces more distinctive region-specific tokenization patterns. We next ask whether these structural improvements translate into downstream predictive performance. Unless otherwise specified, all comparisons use the same model architecture, datasets, and training procedure, with models first pretrained using masked language modeling in a BERT-style setting on DNA sequences from the hg38 human reference genome and then fine-tuned and evaluated on downstream tasks.

Vocabulary Size Selection Based on the token analyses in Section 4, we select **5,120** as the vocabulary size for downstream experiments, as this setting provides the best overall balance between structural fidelity and vocabulary efficiency. Per-vocabulary structural metrics are reported in Tables 2, 3, 6 and 7.

5.1 Controlled Tokenizer Comparison

We compare EvoLen against standard BPE across three benchmark suites: Genome Understanding Evaluation (GUE) (Zhou et al., 2023), Genomic Benchmarks (GBM) (Grešová et al., 2023), and Nucleotide Transformer (NT) (Dalla-Torre et al., 2025). We also evaluate two additional regulatory settings: Multi-SCREEN, a multiclass cis-regulatory element (cCRE) classification task (ENCODE Project Consortium et al., 2020), and snATAC-seq, a cross-species brain cell-type cCRE classification task using human training and mouse evaluation (Zemke et al., 2023). We report performance using Matthews correlation coefficient (MCC), averaged within each benchmark task group (Chicco & Jurman, 2020).

Results. Table 1 reports average MCC across all benchmark groups. EvoLen outperforms the baseline on 11 of 15 task groups, with the clearest gains on tasks involving regulatory structure and cross-species generalization: mouse enhancers in GenomicBench (+9.83%), snATAC-seq cross-species brain cell-type prediction (+9.47%), TF binding in GUE (+5.70%), and mouse classification in GUE (+4.07%). Consistent but smaller improvements appear on promoter-300 prediction in GUE (+2.71%) and across NT histone, enhancer, and promoter prediction (+1.88–2.48%). Across the 56 paired tasks, EvoLen improves over baseline BPE on 40 (mean Δ MCC +0.93, median +0.56; bootstrap 95% CI [+0.34, +1.54] over 10,000 resamples; Wilcoxon signed-rank $p = 1.8 \times 10^{-3}$), indicating gains that are modest in magnitude but stable across tasks. These gains are also broadly consistent with the token-level properties identified in Section 4: better motif preservation (P1) aligns with TF-binding improvements (Avsec et al., 2021), stronger region-specific tokenization patterns (P2) align with enhancer and promoter gains, and improved evolutionary alignment (P3) aligns with gains on mouse and cross-species brain cell-type prediction tasks. Although splice sites are evolutionarily conserved, the decline in performance on these tasks in both GUE and NT likely reflects their reliance on highly localized motifs and precise exon–intron boundary signals (Shrikumar et al., 2018), where exact local matching matters more than broader variable-length grouping. Notably, this gap narrows under compute-matched pretraining: at 200k steps the GUE splice deficit closes entirely (+0.1%) and the NT splice deficit shrinks to −0.9% (Table 1), suggesting the limitation is partly a matter of convergence rather than a fixed property of the tokenizer. Invertebrate tasks also decline slightly (−0.39%), and invertebrate and yeast settings likely benefit less because they fall outside the mammalian evolutionary scope used during tokenizer construction. Taken together, these results indicate that EvoLen provides meaningful improvements, with the clearest benefits on regulatory and cross-species tasks that match the functional and evolutionary biases encoded by the tokenizer.

5.2 Tokenizer Strategy Comparison

We further compare EvoLen with tokenization strategies used in prior DNA language models, including subword tokenization (DNABERT2-style) (Zhou et al., 2023), Nucleotide Transformer tokenization (Dalla-Torre et al., 2025), and GROVER tokenization (Sanabria et al., 2024). These comparisons use a shared modeling pipeline. At 100k steps, EvoLen ranks among the top two of the four tokenization strategies on 13 of 15 task groups, despite training for half as long. Extending EvoLen and the baseline to 200k steps gives a compute-matched comparison: EvoLen improves on 12 of 15 groups, and relative to Grover-200K it reaches a 52-task mean MCC of 59.14 vs. 58.44 ($\Delta = +0.70$) and a 13-group mean of 56.95 vs. 55.07 ($\Delta = +1.88$), both excluding the four splice tasks, together with a GBM suite mean of 72.16 vs. 69.11 across all nine GBM tasks ($\Delta = +3.04$, paired Cohen’s $d = 0.37$). EvoLen ranks in the top two among the four tokenization strategies on 94.2% of non-splice tasks (49 of 52) and on 96.9% of regulatory and cross-species tasks (31 of 32). Per-task results across all 56 tasks are reported in Appendix A.6. Throughout, DNABert2, NT and Grover

	Steps	EMP	Mou	GUE				GBM			NT					
				P3	PC	Spl	TF	HR	ME	Inv	His	Enh	Pro	Spl	cCRE	ATAC
<i>Tokenizer Comparison</i>																
DNAbert2	200k	46.2	53.6	76.1	59.1	73.5	59.0	68.7	48.6	69.9	54.5	46.8	72.9	69.6	22.0	8.8
NT	200k	40.4	46.2	79.1	63.4	75.3	53.9	62.4	50.5	67.8	52.2	46.8	73.4	93.9	21.3	9.8
Grover	200k	48.2	60.8	76.3	62.3	79.9	62.4	72.0	47.7	71.4	54.8	47.9	73.5	79.6	23.9	14.5
<i>Baseline Comparison</i>																
Base	100k	46.8	56.5	76.5	64.2	78.5	57.8	69.9	58.1	76.4	53.7	48.7	74.7	69.3	21.9	12.2
EvoLen	100k	46.8	58.8	78.5	63.6	75.2	61.0	70.3	63.8	76.1	55.0	49.6	76.5	67.8	22.1	13.3
Δ%		+0.1	+4.1	+2.7	-1.0	-4.2	+5.7	+0.5	+9.8	-0.4	+2.3	+1.9	+2.5	-2.2	+1.0	+9.5
<i>Compute-Matched Comparison</i>																
Base	200k	46.9	54.5	77.6	64.8	75.5	58.6	69.9	53.8	76.2	53.5	49.7	74.9	69.0	22.7	10.3
EvoLen	200k	47.9	58.9	78.1	64.4	75.6	61.2	70.8	64.2	77.0	55.4	50.0	76.3	68.4	22.5	13.7
Δ%		+2.1	+8.1	+0.6	-0.6	+0.1	+4.4	+1.3	+19.3	+1.0	+3.6	+0.6	+1.9	-0.9	-0.9	+33.0

Table 1: Average MCC (%) across downstream benchmarks. **Bold** = best; underline = second-best, computed across all seven model rows (including both step counts of Base and EvoLen). $\Delta\%$ = relative improvement of EvoLen over baseline BPE. Steps = number of pretraining steps. Fine-tuning hyperparameters are selected per task. Column abbreviations — GUE: EMP = EMP (Yeast), Mou = Mouse, P3 = Promoter-300, PC = Promoter-Core, Spl = Splice, TF = TF binding; GBM: HR = Human regulatory, ME = Mouse enhancers, Inv = Invertebrates; NT: His = Histone marks, Enh = Enhancers, Pro = Promoters, Spl = Splice; cCRE = Multi-SCREEN (Human cCRE); ATAC = snATAC-seq (Human/Mouse brain).

denote the tokenization schemes introduced by those models; in every case we pretrain and fine-tune our own BERT-style model on hg38 using that tokenizer, rather than evaluating the released models themselves.

5.3 Ablation Study

We ablate the three main components of EvoLen at vocabulary size 5,120: three-way evolutionary partitioning, conservation-prioritized merging, and length-aware scoring. Functional and evolutionary token analysis results are reported in Appendix Table 8, and downstream analysis results are provided in Appendix Table 9. Among the tokenizer-construction components, conservation-prioritized merging has the largest effect on structural quality, while evolutionary partitioning provides a smaller but consistent benefit. In downstream evaluation, full EvoLen performs best on nearly all task groups, with snATAC-seq essentially tied with the **No Partition** variant (13.3 vs. 13.4 MCC, a negligible difference relative to the overall difficulty of cross-species generalization at this scale). Overall, these results suggest that most of EvoLen’s gains come from selecting biologically informative tokens during vocabulary construction, while length-aware decoding provides a refinement.

6 Related Work

DNA language models and tokenization. Recent genomic foundation models have shown that large-scale pretraining can learn useful sequence representations from DNA (Ji et al., 2021; Zhou et al., 2023; Nguyen et al., 2024). Existing tokenization strategies largely fall into three categories: fixed-length k -mers, adaptive subword tokenization, and single-nucleotide resolution. Fixed-length approaches like DNABERT (Ji et al., 2021) impose a uniform scale that can fragment regulatory elements, while adaptive subword methods such as DNABERT2 and GROVER (Zhou et al., 2023; Sanabria et al., 2024) remain driven primarily by frequency rather than biological function. At the other extreme, nucleotide-level models such as Evo and HyenaDNA (Nguyen et al., 2024; 2023) avoid fragmentation but sacrifice the efficiency of learned variable-length subwords (Clark et al., 2022). EvoLen addresses this gap.

Biological priors and evolutionary information. Biological sequence function is organized across multiple scales, from short TF-binding motifs to broader regulatory annotations (Rauluseviciute et al., 2024; ENCODE Project Consortium et al., 2020). Evolutionary conservation has long served as a proxy for functional importance, and scores such as phyloP

provide nucleotide-level estimates of selective constraint across species (Pollard et al., 2010; UCSC Genome Browser, 2024b). Recent large-scale sequence models can absorb such signals implicitly through broad phylogenetic training corpora (Nguyen et al., 2024; Brixi et al., 2026). In contrast, EvoLen incorporates conservation explicitly into subword vocabulary construction and decoding. To our knowledge, prior DNA tokenizers have not incorporated evolutionary conservation directly into subword vocabulary design. EvoLen is complementary to both scaling-oriented DNA language models and existing benchmark efforts (Feng et al., 2025; Grešová et al., 2023).

7 Conclusion

Our results demonstrate that tokenization is not a neutral preprocessing step for DNALMs: it introduces a biologically informed inductive bias that shapes how regulatory sequence patterns are represented and learned. By incorporating evolutionary constraint directly into vocabulary construction and decoding, EvoLen provides this bias without requiring changes to model architecture or training objectives. Across token analyses, EvoLen improves motif preservation, evolutionary alignment, and regulatory element-level differentiation, and these gains translate most clearly to downstream improvements on tasks involving regulatory modeling and cross-species generalization. More broadly, for DNALM, structured tokenization offers a lightweight path to better representations, one that complements, rather than competes with, gains from increased model scale.

Limitations and Future Directions. EvoLen’s benefits are strongest when downstream tasks align with the evolutionary scope of the tokenizer, and the gains are less consistent for tasks dominated by exact short-boundary recognition or for species outside the conservation signal used during tokenizer construction. We assess robustness at the task level rather than across pretraining seeds; multi-seed pretraining is left to future work. A promising next step is to extend EvoLen beyond evolutionary conservation by incorporating additional biological priors into token construction, such as motif coverage, GC context, or other sequence-level features. Another important direction is to test whether biologically informed tokenization yields larger benefits in downstream applications such as sequence design and individual genome annotation (Zhou & Troyanskaya, 2015). Variant effect prediction is a particularly direct test, since it probes whether a tokenizer’s units align with functional sequence elements; it is not a standard benchmark for BERT-style DNALMs — the GUE, GBM and NT suites are classification-oriented — and requires methodology distinct from sequence classification, so we leave it to future work.

Disclosure of LLM Usage

In accordance with Policy 1 regarding the use of Large Language Models (LLMs) in research and reviewing, we disclose the following:

- **Research Content Generation:** No LLMs were used to originate research ideas, write original text, or generate data.

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A Appendix

A.1 Additional Motif Preservation Analysis

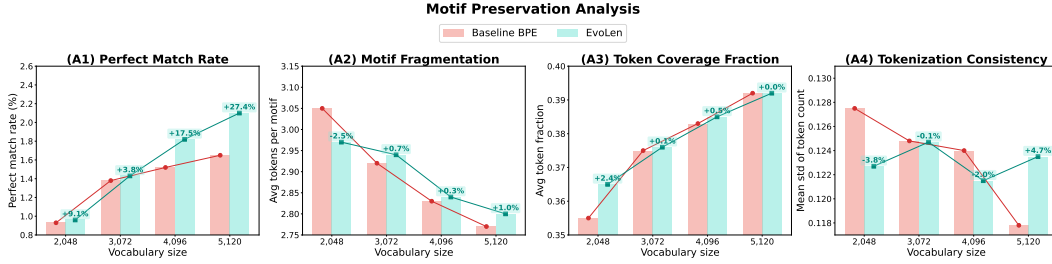


Figure 4: Motif preservation diagnostics across vocabulary sizes: perfect match rate, fragmentation, token coverage fraction, and tokenization consistency. EvoLen improves exact motif preservation while maintaining comparable performance on other metrics.

Since **Motif fragmentation** (average tokens per motif) is heavily influenced by motif length—longer motifs naturally require more tokens regardless of the tokenizer—we report two complementary metrics. **Token Coverage Fraction** normalizes by motif length and measures the average fraction of the motif covered by each token (higher = fewer, longer-spanning tokens). **Tokenization Consistency** measures the standard deviation of token counts across wildcard-expanded variants of the same motif family (lower = more stable segmentation). Formally, for a tokenizer t and motif family m , let $V(m)$ denote the set of all expanded variants of m , and let $n_{m,v}$ denote the number of tokens produced when encoding variant $v \in V(m)$. We define:

$$\text{Consistency}(t) = \frac{1}{|M|} \sum_{m \in M} \text{std}(\{n_{m,v}\}_{v \in V(m)}),$$

where M is the set of motif families. These metrics show that EvoLen’s improvement is concentrated in exact motif preservation rather than bulk fragmentation reduction (Table 2, Figure 4). In Table 2, *AvgTok/Motif* is the average number of tokens a motif is split into; *PerfectMatch%* is the fraction of motifs encoded as a single token; *ExactVocab%* is the fraction of motifs that appear verbatim in the vocabulary; and *AvgTokenFrac* is the average fraction of the motif covered by each token (higher values indicate longer-spanning tokens).

Table 2: Motif preservation summary across tokenizer variants and vocabulary sizes.

Tokenizer	AvgTok/Motif	PerfectMatch%	ExactVocab%	AvgTokenFrac
Baseline BPE 5120	2.776	1.659	1.659	0.389
EvoLen 5120	2.804	2.114	2.114	0.389
Baseline BPE 4096	2.842	1.525	1.525	0.380
EvoLen 4096	2.849	1.793	1.793	0.382
Baseline BPE 3072	2.914	1.391	1.391	0.371
EvoLen 3072	2.934	1.445	1.445	0.371
Baseline BPE 2048	3.049	0.883	0.883	0.355
EvoLen 2048	2.974	0.963	0.963	0.363
Grover	3.404	0.134	0.134	0.322
DNAbert2	2.786	2.435	2.435	0.393

A.2 Evolutionary Conservation Details

Table 3 reports per-region phyloP statistics for both EvoLen and baseline BPE across vocabulary sizes. *MeanPhyloP* is the mean per-token phyloP score averaged across all tokens in a given region; $\%>0$ denotes the percentage of tokens whose mean phyloP is positive, indicating net conservation; *MeanVar* is the mean intra-token phyloP variance, measuring how homogeneous each token’s constituent bases are in conservation level. EvoLen consistently achieves higher mean phyloP in conserved regions, larger improvements in neutral regions, and slightly more negative values in accelerated regions, indicating better alignment with evolutionary constraint.

Motif Functional Integrity: EvoLen vs BPE				
<i>EvoLen recognizes motifs as intact tokens; BPE fragments across motif boundaries (vocab = 5,120)</i>				
TF Name (Motif)		Tokenization — motif region highlighted		
LMX1B (TTAATTAA)	BPE	GC	ATCG	TTAATT AATGC AAC
	EvoLen	GCATC	G	TTAATTAA TGCAAC
NEUROD1 (ACAGATGG)	BPE	GC	ATCG	ACAG ATGG TGCAAC
	EvoLen	GCATC	G	ACAGATGG TGCAAC
Nkx2-1 (TCAAGTG)	BPE	GC	ATCG	TCAAG TGTGC AAC
	EvoLen	GCATC	G	TCAAGTG TGCAAC
LMX1A (TAATTAA)	BPE	GC	ATCG	TAA TTAA TGCAAC
	EvoLen	GCATC	G	TAATTAA TGCAAC
TCF7 (ATCAAAG)	BPE	GC	ATCG	ATC AAAGTGC AAC
	EvoLen	GCATC	G	ATCAAAG TGCAAC
NKX2-5 (TTGAGTG)	BPE	GC	ATCG	TTG AGTG TGCAAC
	EvoLen	GCATC	G	TTGAGTG TGCAAC

Figure 5: Motif fragmentation examples at vocabulary size 5,120. EvoLen preserves motifs as intact tokens, whereas BPE splits them across token boundaries.

Table 3: PhyloP summary by conservation category and vocabulary size, including other tokenization strategies at their native vocabulary sizes.

Method	Vocab	Region	MeanPhyloP	%>0	MeanVar	Tokens
EvoLen	2048	conserved	0.558	100.0	0.283	2,042
EvoLen	2048	neutral	0.080	87.1	0.348	2,042
EvoLen	2048	accelerated	-0.273	0.3	0.635	2,042
Baseline	2048	conserved	0.549	100.0	0.283	2,035
Baseline	2048	neutral	0.072	84.7	0.347	2,042
Baseline	2048	accelerated	-0.271	0.9	0.633	2,042
EvoLen	3072	conserved	0.559	100.0	0.287	3,062
EvoLen	3072	neutral	0.081	87.7	0.352	3,066
EvoLen	3072	accelerated	-0.274	0.3	0.644	3,066
Baseline	3072	conserved	0.546	100.0	0.286	3,041
Baseline	3072	neutral	0.071	84.9	0.352	3,066
Baseline	3072	accelerated	-0.272	0.8	0.642	3,065
EvoLen	4096	conserved	0.557	100.0	0.290	4,083
EvoLen	4096	neutral	0.080	87.3	0.356	4,090
EvoLen	4096	accelerated	-0.275	0.2	0.651	4,090
Baseline	4096	conserved	0.545	100.0	0.289	4,045
Baseline	4096	neutral	0.070	84.6	0.355	4,090
Baseline	4096	accelerated	-0.274	0.7	0.649	4,086
EvoLen	5120	conserved	0.558	100.0	0.291	5,107
EvoLen	5120	neutral	0.081	87.2	0.358	5,114
EvoLen	5120	accelerated	-0.273	0.2	0.654	5,114
Baseline	5120	conserved	0.545	100.0	0.290	5,042
Baseline	5120	neutral	0.070	84.7	0.356	5,109
Baseline	5120	accelerated	-0.272	0.6	0.651	5,106
Grover	605	conserved	0.549	100.0	0.271	604
Grover	605	neutral	0.071	84.8	0.329	604
Grover	605	accelerated	-0.274	1.3	0.601	604
DNAbert2	4096	conserved	0.548	100.0	0.294	4,088
DNAbert2	4096	neutral	0.068	82.1	0.363	4,091
DNAbert2	4096	accelerated	-0.282	0.6	0.656	4,091

A.3 Region-Specific Tokenization Analysis

Table 4: Token length-bin composition (%) across four genomic regions at vocabulary sizes 3,072 and 5,120.

Method	Vocab	Region	Pct1–2	Pct3–5	Pct6–8	Pct9+
Baseline	3072	promoter	4.48	63.72	30.91	0.90
Baseline	3072	enhancer	4.12	62.00	33.23	0.66
Baseline	3072	exon	4.01	62.25	32.91	0.84
Baseline	3072	intron	2.15	61.01	35.32	1.52
EvoLen	3072	promoter	26.59	23.71	49.03	0.67
EvoLen	3072	enhancer	27.79	18.85	52.81	0.55
EvoLen	3072	exon	27.22	20.09	52.03	0.66
EvoLen	3072	intron	26.94	17.20	54.73	1.14
Baseline	5120	promoter	3.27	58.09	37.35	1.29
Baseline	5120	enhancer	3.19	55.98	39.83	1.01
Baseline	5120	exon	3.04	56.31	39.40	1.25
Baseline	5120	intron	1.40	54.35	41.96	2.29
EvoLen	5120	promoter	26.19	17.26	55.20	1.35
EvoLen	5120	enhancer	27.27	14.18	57.34	1.21
EvoLen	5120	exon	26.86	14.90	56.80	1.44
EvoLen	5120	intron	26.35	12.91	58.38	2.36

Tables 4 and 5 report token-length distributions across four genomic regions and pairwise Jensen–Shannon divergences at vocabulary sizes 3,072 and 5,120. Note that the Jensen–Shannon distance reported in the main text (Figure 2B) is the square root of the Jensen–Shannon divergence; the appendix tables report the divergence directly. *Pct1–2*, *Pct3–5*, *Pct6–8*, and *Pct9+* denote the percentage of tokens whose length falls in the corresponding base-pair range. EvoLen produces more distinct length profiles across regions, with a larger fraction of tokens in the 6–8 bp range and greater inter-region separation.

Table 5: Pairwise Jensen–Shannon divergence between region token-length distributions at vocabulary size 5,120.

RegionA	RegionB	Baseline	EvoLen
promoter	enhancer	0.001	0.001
promoter	exon	0.000	0.001
promoter	intron	0.005	0.004
enhancer	exon	0.000	0.000
enhancer	intron	0.005	0.002
exon	intron	0.004	0.001

A.4 Functional Sequence Enrichment Details

To quantify whether a tokenizer captures region- and conservation-specific sequence patterns, we compute the mean $\log_2$ fold-change of per-token frequencies relative to a neutral intronic background. We construct 12 genomic bins by intersecting four region annotations (promoter, enhancer, exon, intron) with three conservation categories (conserved, neutral, accelerated). For a given bin b and the reference background bin bg (intron $\times$ neutral), the smoothed frequency of token t is:

$$f_t^{(b)} = \frac{c_t^{(b)} + \alpha}{N^{(b)} + \alpha \cdot |V|}$$

where $c_t^{(b)}$ is the raw count, $N^{(b)} = \sum_t c_t^{(b)}$, $|V|$ is the vocabulary size, and $\alpha = 0.5$. The per-token log fold-change and enrichment score are:

$$\log_2 \text{FC}(t, b) = \log_2 \frac{f_t^{(b)}}{f_t^{(bg)}}, \quad \bar{E}(b) = \frac{1}{|V|} \sum_{t \in V} \log_2 \text{FC}(t, b)$$

Negative values indicate tokens that are less frequent than in the intronic neutral background; by definition $\bar{E}(\text{intron} \times \text{neutral}) = 0$. Tables 6 and 7 report enrichment diagnostics for promoter-enhancer and exon-intron comparisons.

Table 6: Promoter versus enhancer token enrichment diagnostics.

Model	Vocab	SigTokens	SigProm	SigEnh	PromShare%	EnhMeanLen	EnhMeanGC
Baseline	2048	1,736	456	1,280	26.27	6.21	0.337
EvoLen	2048	1,852	380	1,472	20.52	6.31	0.332
Baseline	3072	2,449	598	1,851	24.42	6.51	0.345
EvoLen	3072	2,746	517	2,229	18.83	6.65	0.336
Baseline	4096	3,062	755	2,307	24.66	6.69	0.347
EvoLen	4096	3,546	685	2,861	19.32	6.86	0.340
Baseline	5120	3,598	895	2,703	24.88	6.84	0.350
EvoLen	5120	4,290	833	3,457	19.42	7.03	0.342

Table 7: Exon versus intron token enrichment summary.

Model	Vocab	SigTokens	SigExon	SigIntron	BestExon	BestIntron
Baseline	2048	1,350	470	880	A	TATT
EvoLen	2048	1,532	558	974	C	T
Baseline	3072	1,754	603	1,151	A	TATT
EvoLen	3072	2,026	720	1,306	C	T
Baseline	4096	2,066	714	1,352	A	TATT
EvoLen	4096	2,442	857	1,585	C	T
Baseline	5120	2,266	786	1,480	A	TATT
EvoLen	5120	2,776	1,025	1,751	C	T

A.5 Ablation Study Details

To isolate the contribution of each component, we design three ablation variants at vocabulary size 5,120: **No Partition** removes three-way genomic partitioning and trains a single whole-genome BPE; **No Priority** removes the conservation-priority merge order; **Linear score** replaces ℓ^2 scoring with linear ℓ . All other aspects remain identical to full EvoLen.

Table 8 reports structural metrics (see Table 2 for metric definitions of Tok/Motif and Perfect%) and Table 9 reports downstream performance.

A.6 Per-Task Downstream Results

Tables 10–14 report per-task baseline vs. EvoLen results across all 56 downstream tasks, and Tables 15–24 report tokenizer-strategy and ablation comparisons. We separate comparisons for readability: baseline vs. EvoLen, tokenizer strategies (DNAbert2, NT, Grover), and ablation variants (w/o Len, w/o Part, w/o Pri). We report Matthews correlation coefficient (MCC), accuracy (Acc), macro F1, precision, and recall. The main-text Table 1 reports group-averaged MCC; these tables provide the full task-level breakdown.

Table 8: Ablation structural analysis (vocab size 5,120). **Bold**: best ablation per metric.

Tokenizer	Mean phyloP	% > 0	Mean Var	Tok/Motif	Perfect%
<i>References</i>					
Baseline BPE	0.082	85.0	0.357	2.78	1.66
EvoLen	0.097	88.7	0.361	2.80	2.11
<i>Ablations</i>					
No Partition	0.085	85.9	0.359	2.78	1.66
No Priority	0.071	81.0	0.374	3.08	1.69
Linear score (ℓ vs ℓ^2)	0.100	88.7	0.359	2.86	2.11

Tokenizer	Region	Mean phyloP	% > 0	Mean Var	Tokens
No Partition	conserved	0.547	100.0	0.291	5,060
No Partition	neutral	0.072	85.2	0.357	5,109
No Partition	accelerated	-0.273	0.5	0.652	5,109
No Priority	conserved	0.539	99.9	0.293	5,006
No Priority	neutral	0.060	80.2	0.367	5,111
No Priority	accelerated	-0.276	0.5	0.661	5,111
Linear score	conserved	0.558	100.0	0.289	5,104
Linear score	neutral	0.083	87.6	0.357	5,114
Linear score	accelerated	-0.270	0.2	0.652	5,114
EvoLen	conserved	0.558	100.0	0.291	5,107
EvoLen	neutral	0.081	87.2	0.358	5,114
EvoLen	accelerated	-0.273	0.2	0.654	5,114

Table 9: Ablation downstream performance: average MCC (%) comparing full EvoLen against variants without length encoding (w/o Len), without partition (w/o Part), and without priority (w/o Pri). **Bold** = best; underline = second best.

Benchmark	Task Group	EvoLen	w/o Len	w/o Part	w/o Pri
GUE	EMP (Yeast)	46.8	45.5	<u>45.7</u>	43.7
GUE	Mouse	58.8	50.2	<u>55.8</u>	51.5
GUE	Promoter-300	78.5	76.7	<u>75.0</u>	<u>77.3</u>
GUE	Promoter-Core	63.6	61.1	<u>63.1</u>	59.2
GUE	Splice	75.2	<u>71.6</u>	69.0	68.9
GUE	TF binding	61.0	58.5	<u>59.0</u>	54.5
GenomicBench	Human regulatory	70.3	68.7	<u>68.8</u>	67.9
GenomicBench	Mouse enhancers	63.8	46.3	47.0	<u>59.2</u>
GenomicBench	Invertebrates	76.1	<u>74.6</u>	74.0	71.1
NT	Histone marks	55.0	<u>54.2</u>	53.7	52.3
NT	Enhancers	49.6	<u>47.4</u>	47.0	45.4
NT	Promoters	76.5	<u>71.9</u>	69.8	70.6
NT	Splice	67.8	<u>66.3</u>	61.2	58.2
Multi-SCREEN	Human cCRE	22.1	21.6	<u>21.6</u>	21.2
snATAC-seq	Human/Mouse brain	<u>13.3</u>	13.0	13.4	12.4

#	Benchmark	Task	Base	EvoLen
1	GBM	demo_coding_vs_intergenomic_seqs	82.2	81.4
2	GBM	demo_human_or_worm	91.8	92.1
3	GBM	drosophila_enhancers_stark	55.3	54.9
4	GBM	dummy_mouse_enhancers_ensembl	58.1	63.8
5	GBM	human_enhancers_cohn	50.2	49.7
6	GBM	human_enhancers_ensembl	77.5	78.3
7	GBM	human_ensembl_regulatory	84.8	84.3
8	GBM	human_nontata_promoters	87.4	89.1
9	GBM	human_ocr_ensembl	49.9	50.0
10	GUE	H3	69.6	73.9
11	GUE	H3K14ac	40.7	40.1
12	GUE	H3K36me3	42.8	44.5
13	GUE	H3K4me1	41.2	36.3
14	GUE	H3K4me2	30.5	30.4
15	GUE	H3K4me3	25.5	26.1
16	GUE	H3K79me3	58.1	59.0
17	GUE	H3K9ac	49.1	49.3
18	GUE	H4	72.7	73.9
19	GUE	H4ac	37.6	34.6
20	GUE	mouse_0	40.0	43.1
21	GUE	mouse_1	79.7	79.1
22	GUE	mouse_2	74.6	75.7
23	GUE	mouse_3	55.6	60.7
24	GUE	mouse_4	32.4	35.2
25	GUE	300_all	82.5	82.7
26	GUE	300_notata	89.5	90.7
27	GUE	300_tata	57.4	62.1
28	GUE	core_all	62.3	62.7
29	GUE	core_notata	65.8	66.0
30	GUE	core_tata	64.6	62.2
31	GUE	splice_reconstructed	78.5	75.2
32	GUE	tf_0	67.2	65.5
33	GUE	tf_1	64.9	67.8
34	GUE	tf_2	52.1	57.2
35	GUE	tf_3	40.2	45.7
36	GUE	tf_4	64.2	69.1
37	snATAC-seq	Human/Mouse brain	12.2	13.3
38	Multi-SCREEN	Human cCRE	21.9	22.1
39	NT	H2AFZ	48.1	49.3
40	NT	H3K27ac	46.5	46.0
41	NT	H3K27me3	58.1	58.6
42	NT	H3K36me3	62.1	61.7
43	NT	H3K4me1	45.2	47.8
44	NT	H3K4me2	54.7	58.5
45	NT	H3K4me3	64.7	65.1
46	NT	H3K9ac	53.8	54.6
47	NT	H3K9me3	38.9	42.0
48	NT	H4K20me1	65.2	66.0
49	NT	enhancers	51.1	52.8
50	NT	enhancers_types	46.3	46.5
51	NT	promoter_all	73.9	74.4
52	NT	promoter_no_tata	76.2	77.5
53	NT	promoter_tata	73.9	77.7
54	NT	splice_sites_acceptors	66.9	66.6
55	NT	splice_sites_all	67.4	68.0
56	NT	splice_sites_donors	73.6	68.9

Table 10: Per-task MCC (%) for baseline vs EvoLen comparison.

#	Benchmark	Task	Base	EvoLen
1	GBM	demo_coding_vs_intergenomic_seqs	91.1	90.7
2	GBM	demo_human_or_worm	95.9	96.1
3	GBM	drosophila_enhancers_stark	77.8	77.6
4	GBM	dummy_mouse_enhancers_ensembl	79.3	81.8
5	GBM	human_enhancers_cohn	75.1	74.9
6	GBM	human_enhancers_ensembl	88.7	89.1
7	GBM	human_ensembl_regulatory	89.8	89.5
8	GBM	human_nontata_promoters	93.5	94.5
9	GBM	human_ocr_ensembl	74.9	75.0
10	GUE	H3	84.7	87.0
11	GUE	H3K14ac	71.0	70.6
12	GUE	H3K36me3	71.4	72.6
13	GUE	H3K4me1	71.0	68.3
14	GUE	H3K4me2	67.2	67.1
15	GUE	H3K4me3	62.9	63.1
16	GUE	H3K79me3	79.1	79.5
17	GUE	H3K9ac	74.8	74.2
18	GUE	H4	86.5	87.1
19	GUE	H4ac	68.9	67.4
20	GUE	mouse_0	70.0	71.5
21	GUE	mouse_1	89.8	89.5
22	GUE	mouse_2	87.2	87.8
23	GUE	mouse_3	77.8	80.3
24	GUE	mouse_4	66.2	67.6
25	GUE	300_all	91.2	91.4
26	GUE	300_notata	94.7	95.3
27	GUE	300_tata	78.6	81.1
28	GUE	core_all	81.2	81.3
29	GUE	core_notata	82.9	83.0
30	GUE	core_tata	82.1	81.1
31	GUE	splice_reconstructed	87.4	85.5
32	GUE	tf_0	83.4	82.3
33	GUE	tf_1	81.0	83.6
34	GUE	tf_2	75.8	78.5
35	GUE	tf_3	69.1	72.6
36	GUE	tf_4	82.1	84.5
37	snATAC-seq	Human/Mouse brain	38.2	39.1
38	Multi-SCREEN	Human cCRE	31.5	31.6
39	NT	H2AFZ	73.9	74.1
40	NT	H3K27ac	73.3	72.9
41	NT	H3K27me3	78.9	79.1
42	NT	H3K36me3	81.0	80.3
43	NT	H3K4me1	72.3	73.0
44	NT	H3K4me2	77.4	79.0
45	NT	H3K4me3	82.1	82.4
46	NT	H3K9ac	76.9	77.2
47	NT	H3K9me3	69.4	71.0
48	NT	H4K20me1	82.5	83.0
49	NT	enhancers	75.5	75.9
50	NT	enhancers_types	71.7	70.9
51	NT	promoter_all	86.9	87.2
52	NT	promoter_no_tata	88.0	88.7
53	NT	promoter_tata	86.9	88.8
54	NT	splice_sites_acceptors	83.4	83.3
55	NT	splice_sites_all	78.3	78.6
56	NT	splice_sites_donors	86.8	84.4

Table 11: Per-task ACC (%) for baseline vs EvoLen comparison.

#	Benchmark	Task	Base	EvoLen
1	GBM	demo_coding_vs_intergenomic_seqs	91.1	90.7
2	GBM	demo_human_or_worm	95.9	96.1
3	GBM	drosophila_enhancers_stark	77.3	77.4
4	GBM	dummy_mouse_enhancers_ensembl	79.1	81.7
5	GBM	human_enhancers_cohn	75.1	74.9
6	GBM	human_enhancers_ensembl	88.6	89.0
7	GBM	human_ensembl_regulatory	90.0	89.6
8	GBM	human_nontata_promoters	93.5	94.5
9	GBM	human_ocr_ensembl	74.9	75.0
10	GUE	H3	84.7	87.0
11	GUE	H3K14ac	70.3	70.0
12	GUE	H3K36me3	71.3	72.2
13	GUE	H3K4me1	70.5	68.1
14	GUE	H3K4me2	64.0	64.5
15	GUE	H3K4me3	62.8	63.0
16	GUE	H3K79me3	79.0	79.4
17	GUE	H3K9ac	74.6	74.2
18	GUE	H4	86.0	86.9
19	GUE	H4ac	68.8	67.3
20	GUE	mouse_0	70.0	71.4
21	GUE	mouse_1	89.8	89.5
22	GUE	mouse_2	87.2	87.8
23	GUE	mouse_3	77.8	80.3
24	GUE	mouse_4	66.1	67.6
25	GUE	300_all	91.2	91.3
26	GUE	300_notata	94.7	95.3
27	GUE	300_tata	78.6	81.0
28	GUE	core_all	81.2	81.3
29	GUE	core_notata	82.9	83.0
30	GUE	core_tata	82.0	81.1
31	GUE	splice_reconstructed	86.6	84.5
32	GUE	tf_0	83.3	82.2
33	GUE	tf_1	80.6	83.5
34	GUE	tf_2	75.7	78.5
35	GUE	tf_3	68.3	72.4
36	GUE	tf_4	82.1	84.5
37	snATAC-seq	Human/Mouse brain	27.1	32.0
38	Multi-SCREEN	Human cCRE	30.4	30.6
39	NT	H2AFZ	73.8	73.7
40	NT	H3K27ac	73.3	72.8
41	NT	H3K27me3	78.9	79.1
42	NT	H3K36me3	81.0	80.2
43	NT	H3K4me1	72.2	72.7
44	NT	H3K4me2	77.3	78.7
45	NT	H3K4me3	82.0	82.4
46	NT	H3K9ac	76.9	77.2
47	NT	H3K9me3	69.4	71.0
48	NT	H4K20me1	82.5	83.0
49	NT	enhancers	75.5	75.7
50	NT	enhancers_types	48.8	55.4
51	NT	promoter_all	86.9	87.1
52	NT	promoter_no_tata	88.0	88.7
53	NT	promoter_tata	86.9	88.8
54	NT	splice_sites_acceptors	83.4	83.3
55	NT	splice_sites_all	78.3	78.6
56	NT	splice_sites_donors	86.8	84.4

Table 12: Per-task F1 (%) for baseline vs EvoLen comparison.

#	Benchmark	Task	Base	EvoLen
1	GBM	demo_coding_vs_intergenomic_seqs	91.1	90.7
2	GBM	demo_human_or_worm	95.9	96.1
3	GBM	drosophila_enhancers_stark	78.2	77.5
4	GBM	dummy_mouse_enhancers_ensembl	79.0	81.7
5	GBM	human_enhancers_cohn	75.1	74.9
6	GBM	human_enhancers_ensembl	88.9	89.3
7	GBM	human_ensembl_regulatory	90.4	90.2
8	GBM	human_nontata_promoters	93.5	94.4
9	GBM	human_ocr_ensembl	74.9	75.0
10	GUE	H3	84.9	87.0
11	GUE	H3K14ac	70.4	70.0
12	GUE	H3K36me3	71.3	72.2
13	GUE	H3K4me1	70.8	68.1
14	GUE	H3K4me2	66.6	66.1
15	GUE	H3K4me3	62.8	63.0
16	GUE	H3K79me3	79.1	79.4
17	GUE	H3K9ac	74.5	74.5
18	GUE	H4	87.1	87.1
19	GUE	H4ac	68.8	67.3
20	GUE	mouse_0	70.0	71.6
21	GUE	mouse_1	89.8	89.5
22	GUE	mouse_2	87.4	87.9
23	GUE	mouse_3	77.8	80.4
24	GUE	mouse_4	66.2	67.6
25	GUE	300_all	91.2	91.4
26	GUE	300_notata	94.8	95.3
27	GUE	300_tata	78.7	81.1
28	GUE	core_all	81.2	81.4
29	GUE	core_notata	82.9	83.0
30	GUE	core_tata	82.5	81.1
31	GUE	splice_reconstructed	86.2	84.6
32	GUE	tf_0	83.9	83.2
33	GUE	tf_1	84.0	84.2
34	GUE	tf_2	76.3	78.7
35	GUE	tf_3	71.2	73.1
36	GUE	tf_4	82.2	84.6
37	snATAC-seq	Human/Mouse brain	36.4	35.4
38	Multi-SCREEN	Human cCRE	30.7	31.2
39	NT	H2AFZ	74.3	75.4
40	NT	H3K27ac	73.3	73.1
41	NT	H3K27me3	79.2	79.4
42	NT	H3K36me3	81.1	81.3
43	NT	H3K4me1	72.8	74.6
44	NT	H3K4me2	77.4	79.8
45	NT	H3K4me3	82.6	82.7
46	NT	H3K9ac	76.9	77.3
47	NT	H3K9me3	69.5	71.0
48	NT	H4K20me1	82.7	83.0
49	NT	enhancers	75.5	76.7
50	NT	enhancers_types	48.0	65.4
51	NT	promoter_all	86.9	87.2
52	NT	promoter_no_tata	88.3	88.8
53	NT	promoter_tata	86.9	88.9
54	NT	splice_sites_acceptors	83.4	83.3
55	NT	splice_sites_all	78.3	78.6
56	NT	splice_sites_donors	86.8	84.5

Table 13: Per-task Precision (%) for baseline vs EvoLen comparison.

#	Benchmark	Task	Base	EvoLen
1	GBM	demo_coding_vs_intergenomic_seqs	91.1	90.7
2	GBM	demo_human_or_worm	95.9	96.1
3	GBM	drosophila_enhancers_stark	77.1	77.3
4	GBM	dummy_mouse_enhancers_ensembl	79.1	82.2
5	GBM	human_enhancers_cohn	75.1	74.9
6	GBM	human_enhancers_ensembl	88.6	89.0
7	GBM	human_ensembl_regulatory	89.7	89.3
8	GBM	human_nontata_promoters	93.9	94.7
9	GBM	human_ocr_ensembl	74.9	75.0
10	GUE	H3	84.7	87.0
11	GUE	H3K14ac	70.3	70.2
12	GUE	H3K36me3	71.5	72.3
13	GUE	H3K4me1	70.5	68.2
14	GUE	H3K4me2	64.0	64.3
15	GUE	H3K4me3	62.7	63.1
16	GUE	H3K79me3	79.0	79.5
17	GUE	H3K9ac	74.6	74.7
18	GUE	H4	85.6	86.8
19	GUE	H4ac	68.8	67.3
20	GUE	mouse_0	70.0	71.5
21	GUE	mouse_1	89.8	89.5
22	GUE	mouse_2	87.2	87.8
23	GUE	mouse_3	77.8	80.3
24	GUE	mouse_4	66.2	67.6
25	GUE	300_all	91.2	91.3
26	GUE	300_notata	94.7	95.3
27	GUE	300_tata	78.7	81.0
28	GUE	core_all	81.2	81.3
29	GUE	core_notata	82.9	83.0
30	GUE	core_tata	82.1	81.1
31	GUE	splice_reconstructed	86.9	84.4
32	GUE	tf_0	83.4	82.3
33	GUE	tf_1	81.0	83.6
34	GUE	tf_2	75.8	78.5
35	GUE	tf_3	69.1	72.6
36	GUE	tf_4	82.1	84.5
37	snATAC-seq	Human/Mouse brain	31.6	33.1
38	Multi-SCREEN	Human cCRE	31.6	31.7
39	NT	H2AFZ	73.8	73.9
40	NT	H3K27ac	73.3	72.9
41	NT	H3K27me3	79.0	79.2
42	NT	H3K36me3	81.0	80.4
43	NT	H3K4me1	72.4	73.2
44	NT	H3K4me2	77.3	78.7
45	NT	H3K4me3	82.1	82.4
46	NT	H3K9ac	76.9	77.2
47	NT	H3K9me3	69.4	71.0
48	NT	H4K20me1	82.6	83.0
49	NT	enhancers	75.5	76.0
50	NT	enhancers_types	50.0	54.0
51	NT	promoter_all	86.9	87.1
52	NT	promoter_no_tata	88.0	88.6
53	NT	promoter_tata	86.9	88.8
54	NT	splice_sites_acceptors	83.5	83.3
55	NT	splice_sites_all	78.4	78.7
56	NT	splice_sites_donors	86.8	84.4

Table 14: Per-task Recall (%) for baseline vs EvoLen comparison.

#	Benchmark	Task	DNAbert2	NT	Grover
1	GBM	demo_coding_vs_intergenomic_seqs	80.5	77.9	83.1
2	GBM	demo_human_or_worm	91.6	87.7	92.8
3	GBM	drosophila_enhancers_stark	37.5	37.9	38.4
4	GBM	dummy_mouse_enhancers_ensembl	48.6	50.5	47.7
5	GBM	human_enhancers_cohn	45.9	43.5	47.4
6	GBM	human_enhancers_ensembl	78.7	66.9	81.1
7	GBM	human_ensembl_regulatory	84.9	86.3	85.5
8	GBM	human_nontata_promoters	86.3	71.1	90.5
9	GBM	human_ocr_ensembl	47.7	44.1	55.5
10	GUE	H3	71.1	65.9	74.9
11	GUE	H3K14ac	37.8	30.4	41.3
12	GUE	H3K36me3	42.6	36.3	44.6
13	GUE	H3K4me1	37.6	34.1	38.1
14	GUE	H3K4me2	30.2	29.5	30.2
15	GUE	H3K4me3	26.2	17.0	29.9
16	GUE	H3K79me3	58.1	52.0	59.1
17	GUE	H3K9ac	49.3	44.2	51.4
18	GUE	H4	75.7	66.9	72.9
19	GUE	H4ac	34.0	27.4	39.7
20	GUE	mouse_0	32.8	30.7	43.7
21	GUE	mouse_1	78.9	72.0	80.5
22	GUE	mouse_2	75.0	62.3	72.7
23	GUE	mouse_3	49.9	44.9	66.0
24	GUE	mouse_4	31.2	21.2	41.1
25	GUE	prom_300_all	81.0	84.0	82.5
26	GUE	prom_300_notata	90.1	89.1	89.8
27	GUE	prom_300_tata	57.3	64.1	56.7
28	GUE	prom_core_all	61.0	60.5	61.4
29	GUE	prom_core_notata	64.5	62.5	65.6
30	GUE	prom_core_tata	51.8	67.4	59.9
31	GUE	splice_reconstructed	73.5	75.3	79.9
32	GUE	tf_0	64.2	61.3	67.1
33	GUE	tf_1	69.3	64.6	68.1
34	GUE	tf_2	55.8	45.9	59.8
35	GUE	tf_3	37.7	34.7	45.3
36	GUE	tf_4	68.1	63.0	71.6
37	snATAC-seq	Human/Mouse brain	8.8	9.8	14.5
38	Multi-SCREEN	Human cCRE	22.0	21.3	23.9
39	NT	H2AFZ	47.0	47.9	47.2
40	NT	H3K27ac	46.2	42.3	47.0
41	NT	H3K27me3	60.0	60.7	61.1
42	NT	H3K36me3	60.4	57.4	60.3
43	NT	H3K4me1	47.6	47.7	48.4
44	NT	H3K4me2	56.8	55.8	56.6
45	NT	H3K4me3	66.4	65.6	67.7
46	NT	H3K9ac	53.4	47.6	54.5
47	NT	H3K9me3	44.1	36.3	42.6
48	NT	H4K20me1	62.8	60.9	62.9
49	NT	enhancers	47.7	49.6	50.1
50	NT	enhancers_types	46.0	43.9	45.8
51	NT	promoter_all	73.4	71.8	73.9
52	NT	promoter_no_tata	73.6	75.3	74.7
53	NT	promoter_tata	71.6	72.9	72.0
54	NT	splice_sites_acceptors	71.2	93.1	74.9
55	NT	splice_sites_all	69.5	93.6	80.8
56	NT	splice_sites_donors	67.9	95.0	82.9

Table 15: Per-task MCC (%) for tokenizer-strategy comparison.

#	Benchmark	Task	w/o Len	w/o Part	w/o Pri
1	GBM	demo_coding_vs_intergenomic_seqs	80.9	81.1	80.2
2	GBM	demo_human_or_worm	90.5	91.7	91.2
3	GBM	drosophila_enhancers_stark	52.4	49.2	41.8
4	GBM	dummy_mouse_enhancers_ensembl	46.3	47.0	59.2
5	GBM	human_enhancers_cohn	48.7	46.5	47.3
6	GBM	human_enhancers_ensembl	76.9	77.3	75.4
7	GBM	human_ensembl_regulatory	84.4	83.5	83.4
8	GBM	human_nontata_promoters	85.8	87.5	86.1
9	GBM	human_ocr_ensembl	48.0	49.3	47.1
10	GUE	H3	69.8	70.0	66.6
11	GUE	H3K14ac	38.7	39.0	34.7
12	GUE	H3K36me3	42.3	42.8	40.7
13	GUE	H3K4me1	37.0	38.6	36.3
14	GUE	H3K4me2	30.9	29.2	28.3
15	GUE	H3K4me3	25.5	23.8	25.7
16	GUE	H3K79me3	57.3	59.8	55.3
17	GUE	H3K9ac	46.4	47.2	44.7
18	GUE	H4	72.9	72.7	70.9
19	GUE	H4ac	34.1	34.0	33.6
20	GUE	mouse_0	33.3	36.2	28.6
21	GUE	mouse_1	77.2	77.6	76.0
22	GUE	mouse_2	64.1	65.9	68.4
23	GUE	mouse_3	44.0	68.3	54.0
24	GUE	mouse_4	32.5	31.1	30.4
25	GUE	prom_300_all	82.3	82.4	81.5
26	GUE	prom_300_notata	89.1	90.8	89.9
27	GUE	prom_300_tata	58.8	51.9	60.5
28	GUE	prom_core_all	61.9	61.7	60.0
29	GUE	prom_core_notata	64.8	65.4	64.6
30	GUE	prom_core_tata	56.4	62.1	53.0
31	GUE	splice_reconstructed	71.6	69.0	68.9
32	GUE	tf_0	66.4	64.0	62.0
33	GUE	tf_1	66.2	68.1	64.9
34	GUE	tf_2	52.9	54.6	46.6
35	GUE	tf_3	42.2	42.4	36.3
36	GUE	tf_4	64.8	65.9	62.6
37	snATAC-seq	Human/Mouse brain	13.0	13.4	12.4
38	Multi-SCREEN	Human cCRE	21.6	21.6	21.2
39	NT	H2AFZ	47.7	45.7	44.3
40	NT	H3K27ac	46.2	43.8	42.4
41	NT	H3K27me3	60.2	61.3	60.3
42	NT	H3K36me3	58.9	56.7	57.1
43	NT	H3K4me1	48.7	48.5	46.8
44	NT	H3K4me2	57.0	57.1	54.5
45	NT	H3K4me3	66.6	66.4	65.1
46	NT	H3K9ac	52.9	53.0	49.0
47	NT	H3K9me3	40.7	41.6	41.4
48	NT	H4K20me1	62.6	63.3	62.3
49	NT	enhancers	48.0	47.7	46.2
50	NT	enhancers_types	46.7	46.3	44.5
51	NT	promoter_all	72.9	71.2	69.9
52	NT	promoter_no_tata	74.3	74.1	73.7
53	NT	promoter_tata	68.6	64.1	68.2
54	NT	splice_sites_acceptors	67.3	63.1	62.0
55	NT	splice_sites_all	64.6	59.3	55.5
56	NT	splice_sites_donors	66.9	61.2	57.1

Table 16: Per-task MCC (%) for ablation variants.

#	Benchmark	Task	DNAbert2	NT	Grover
1	GBM	demo_coding_vs_intergenomic_seqs	90.2	89.0	91.5
2	GBM	demo_human_or_worm	95.8	93.8	96.4
3	GBM	drosophila_enhancers_stark	69.1	69.1	69.5
4	GBM	dummy_mouse_enhancers_ensembl	71.9	75.2	72.7
5	GBM	human_enhancers_cohn	72.8	71.7	73.6
6	GBM	human_enhancers_ensembl	89.3	83.4	90.5
7	GBM	human_ensembl_regulatory	89.9	90.8	90.3
8	GBM	human_nontata_promoters	93.0	85.7	95.1
9	GBM	human_ocr_ensembl	73.8	72.0	77.7
10	GUE	H3	85.5	82.9	87.4
11	GUE	H3K14ac	69.7	66.5	71.5
12	GUE	H3K36me3	71.6	68.4	72.5
13	GUE	H3K4me1	69.2	67.6	69.4
14	GUE	H3K4me2	67.1	66.7	67.1
15	GUE	H3K4me3	63.3	59.0	65.0
16	GUE	H3K79me3	79.1	76.1	79.6
17	GUE	H3K9ac	74.7	72.1	76.0
18	GUE	H4	88.0	83.7	86.6
19	GUE	H4ac	67.0	64.0	70.1
20	GUE	mouse_0	66.4	65.3	71.9
21	GUE	mouse_1	89.4	85.9	90.2
22	GUE	mouse_2	87.5	81.1	86.3
23	GUE	mouse_3	74.9	72.4	82.8
24	GUE	mouse_4	65.6	60.6	70.5
25	GUE	prom_300_all	90.5	92.0	91.2
26	GUE	prom_300_notata	95.1	94.6	94.9
27	GUE	prom_300_tata	78.6	82.1	78.3
28	GUE	prom_core_all	80.5	80.2	80.7
29	GUE	prom_core_notata	82.2	81.2	82.8
30	GUE	prom_core_tata	75.9	83.7	79.9
31	GUE	splice_reconstructed	84.3	85.1	88.2
32	GUE	tf_0	81.7	80.3	83.2
33	GUE	tf_1	84.5	81.9	83.7
34	GUE	tf_2	77.5	72.7	79.0
35	GUE	tf_3	68.2	67.1	72.2
36	GUE	tf_4	84.0	81.4	85.8
37	snATAC-seq	Human/Mouse brain	35.3	37.2	39.6
38	Multi-SCREEN	Human cCRE	31.4	30.9	33.2
39	NT	H2AFZ	73.5	73.4	73.3
40	NT	H3K27ac	73.1	71.0	73.3
41	NT	H3K27me3	80.0	80.0	80.4
42	NT	H3K36me3	80.0	78.5	80.1
43	NT	H3K4me1	73.7	72.8	73.8
44	NT	H3K4me2	78.3	77.5	78.2
45	NT	H3K4me3	83.1	82.8	83.5
46	NT	H3K9ac	76.7	73.7	77.2
47	NT	H3K9me3	72.0	68.1	71.3
48	NT	H4K20me1	81.4	80.4	81.4
49	NT	enhancers	73.8	74.6	75.1
50	NT	enhancers_types	71.6	70.3	71.5
51	NT	promoter_all	86.7	85.9	86.9
52	NT	promoter_no_tata	86.8	87.6	87.3
53	NT	promoter_tata	85.8	86.4	86.0
54	NT	splice_sites_acceptors	85.6	96.5	87.4
55	NT	splice_sites_all	79.7	95.8	87.2
56	NT	splice_sites_donors	84.0	97.5	91.5

Table 17: Per-task Acc (%) for tokenizer-strategy comparison.

#	Benchmark	Task	w/o Len	w/o Part	w/o Pri
1	GBM	demo_coding_vs_intergenomic_seqs	90.4	90.6	90.1
2	GBM	demo_human_or_worm	95.2	95.8	95.6
3	GBM	drosophila_enhancers_stark	76.2	74.7	71.0
4	GBM	dummy_mouse_enhancers_ensembl	73.6	71.9	78.5
5	GBM	human_enhancers_cohn	74.4	73.2	73.6
6	GBM	human_enhancers_ensembl	88.4	88.6	87.6
7	GBM	human_ensembl_regulatory	89.6	89.0	88.9
8	GBM	human_nontata_promoters	92.7	93.6	92.9
9	GBM	human_ocr_ensembl	73.9	74.6	73.5
10	GUE	H3	84.8	84.9	83.3
11	GUE	H3K14ac	70.3	69.6	67.7
12	GUE	H3K36me3	71.5	71.5	70.8
13	GUE	H3K4me1	69.0	69.7	68.6
14	GUE	H3K4me2	67.4	66.7	66.3
15	GUE	H3K4me3	63.0	62.2	63.2
16	GUE	H3K79me3	78.7	80.0	77.7
17	GUE	H3K9ac	72.4	73.7	72.2
18	GUE	H4	86.7	86.5	85.7
19	GUE	H4ac	67.3	67.3	67.0
20	GUE	mouse_0	66.7	68.0	64.3
21	GUE	mouse_1	88.6	88.8	88.0
22	GUE	mouse_2	82.0	82.9	84.1
23	GUE	mouse_3	72.0	84.1	77.0
24	GUE	mouse_4	66.1	65.5	65.1
25	GUE	prom_300_all	91.2	91.2	90.7
26	GUE	prom_300_notata	94.5	95.4	94.9
27	GUE	prom_300_tata	79.1	75.7	80.1
28	GUE	prom_core_all	80.9	80.9	80.0
29	GUE	prom_core_notata	82.4	82.7	82.3
30	GUE	prom_core_tata	77.8	80.8	76.5
31	GUE	splice_reconstructed	83.4	81.9	81.9
32	GUE	tf_0	83.0	81.8	80.7
33	GUE	tf_1	82.6	83.7	82.0
34	GUE	tf_2	75.8	77.3	73.0
35	GUE	tf_3	71.0	70.4	67.5
36	GUE	tf_4	82.4	82.9	81.3
37	snATAC-seq	Human/Mouse brain	38.3	38.8	38.3
38	Multi-SCREEN	Human cCRE	31.1	31.2	30.8
39	NT	H2AFZ	73.8	72.0	72.0
40	NT	H3K27ac	72.8	71.7	71.0
41	NT	H3K27me3	80.0	80.6	79.5
42	NT	H3K36me3	79.3	78.3	78.5
43	NT	H3K4me1	74.2	73.9	72.9
44	NT	H3K4me2	78.4	78.4	77.0
45	NT	H3K4me3	83.3	83.2	82.6
46	NT	H3K9ac	76.5	76.5	74.4
47	NT	H3K9me3	70.4	70.8	70.7
48	NT	H4K20me1	81.3	81.7	81.1
49	NT	enhancers	74.0	73.9	73.1
50	NT	enhancers_types	71.9	71.8	70.9
51	NT	promoter_all	86.1	85.6	85.0
52	NT	promoter_no_tata	87.1	87.0	86.7
53	NT	promoter_tata	84.3	82.0	84.1
54	NT	splice_sites_acceptors	83.7	81.5	81.0
55	NT	splice_sites_all	76.4	72.8	70.2
56	NT	splice_sites_donors	83.5	80.6	78.5

Table 18: Per-task Acc (%) for ablation variants.

#	Benchmark	Task	DNAbert2	NT	Grover
1	GBM	demo_coding_vs_intergenomic_seqs	90.2	89.0	91.5
2	GBM	demo_human_or_worm	95.8	93.8	96.4
3	GBM	drosophila_enhancers_stark	68.5	68.9	68.8
4	GBM	dummy_mouse_enhancers_ensembl	71.8	75.1	72.7
5	GBM	human_enhancers_cohn	72.8	71.7	73.6
6	GBM	human_enhancers_ensembl	89.3	83.4	90.5
7	GBM	human_ensembl_regulatory	90.1	90.9	90.5
8	GBM	human_nontata_promoters	93.0	85.5	95.1
9	GBM	human_ocr_ensembl	73.8	71.9	77.7
10	GUE	H3	85.5	82.9	87.4
11	GUE	H3K14ac	68.8	64.9	70.6
12	GUE	H3K36me3	71.3	68.1	72.3
13	GUE	H3K4me1	68.8	66.8	69.0
14	GUE	H3K4me2	63.8	63.8	63.9
15	GUE	H3K4me3	63.1	57.7	64.9
16	GUE	H3K79me3	79.0	75.9	79.5
17	GUE	H3K9ac	74.5	72.0	75.7
18	GUE	H4	87.8	83.4	86.2
19	GUE	H4ac	66.9	63.4	69.8
20	GUE	mouse_0	66.4	65.3	71.8
21	GUE	mouse_1	89.4	85.9	90.2
22	GUE	mouse_2	87.5	81.1	86.3
23	GUE	mouse_3	74.8	72.4	82.8
24	GUE	mouse_4	65.6	60.6	70.5
25	GUE	prom_300_all	90.5	92.0	91.2
26	GUE	prom_300_notata	95.1	94.6	94.9
27	GUE	prom_300_tata	78.6	82.0	78.3
28	GUE	prom_core_all	80.5	80.2	80.7
29	GUE	prom_core_notata	82.2	81.2	82.8
30	GUE	prom_core_tata	75.8	83.7	79.9
31	GUE	splice_reconstructed	83.4	84.3	87.4
32	GUE	tf_0	81.6	80.2	83.1
33	GUE	tf_1	84.5	81.8	83.6
34	GUE	tf_2	77.3	72.5	78.7
35	GUE	tf_3	67.7	66.8	71.9
36	GUE	tf_4	84.0	81.4	85.8
37	snATAC-seq	Human/Mouse brain	28.6	28.8	32.0
38	Multi-SCREEN	Human cCRE	30.1	29.7	32.1
39	NT	H2AFZ	73.5	73.1	73.3
40	NT	H3K27ac	73.1	71.0	73.3
41	NT	H3K27me3	80.0	79.9	80.4
42	NT	H3K36me3	80.0	78.4	80.1
43	NT	H3K4me1	73.7	72.4	73.7
44	NT	H3K4me2	78.2	77.2	78.2
45	NT	H3K4me3	83.1	82.8	83.5
46	NT	H3K9ac	76.7	73.5	77.2
47	NT	H3K9me3	71.9	67.7	71.3
48	NT	H4K20me1	81.4	80.3	81.4
49	NT	enhancers	73.8	74.5	75.0
50	NT	enhancers_types	50.6	47.8	49.2
51	NT	promoter_all	86.7	85.9	86.9
52	NT	promoter_no_tata	86.8	87.6	87.3
53	NT	promoter_tata	85.8	86.4	86.0
54	NT	splice_sites_acceptors	85.6	96.5	87.4
55	NT	splice_sites_all	79.7	95.7	87.2
56	NT	splice_sites_donors	84.0	97.5	91.4

Table 19: Per-task F1 (%) for tokenizer-strategy comparison.

#	Benchmark	Task	w/o Len	w/o Part	w/o Pri
1	GBM	demo_coding_vs_intergenomic_seqs	90.4	90.6	90.1
2	GBM	demo_human_or_worm	95.2	95.8	95.6
3	GBM	drosophila_enhancers_stark	76.1	74.6	70.9
4	GBM	dummy_mouse_enhancers_ensembl	73.1	71.9	78.5
5	GBM	human_enhancers_cohn	74.4	73.2	73.6
6	GBM	human_enhancers_ensembl	88.4	88.6	87.6
7	GBM	human_ensembl_regulatory	89.7	89.1	89.0
8	GBM	human_nontata_promoters	92.7	93.6	92.9
9	GBM	human_ocr_ensembl	73.9	74.6	73.5
10	GUE	H3	84.8	84.9	83.3
11	GUE	H3K14ac	69.3	69.3	67.3
12	GUE	H3K36me3	71.2	71.3	70.3
13	GUE	H3K4me1	68.4	69.2	68.1
14	GUE	H3K4me2	64.1	63.0	62.5
15	GUE	H3K4me3	62.5	61.5	62.7
16	GUE	H3K79me3	78.7	79.9	77.6
17	GUE	H3K9ac	72.4	73.5	72.1
18	GUE	H4	86.4	86.3	85.4
19	GUE	H4ac	67.0	66.8	66.8
20	GUE	mouse_0	66.7	68.0	64.3
21	GUE	mouse_1	88.6	88.8	88.0
22	GUE	mouse_2	82.0	82.9	84.1
23	GUE	mouse_3	71.9	84.1	77.0
24	GUE	mouse_4	66.0	65.4	65.0
25	GUE	prom_300_all	91.2	91.2	90.7
26	GUE	prom_300_notata	94.5	95.4	94.9
27	GUE	prom_300_tata	79.1	75.7	80.1
28	GUE	prom_core_all	80.9	80.9	80.0
29	GUE	prom_core_notata	82.4	82.7	82.3
30	GUE	prom_core_tata	77.7	80.7	76.5
31	GUE	splice_reconstructed	82.2	80.6	80.6
32	GUE	tf_0	83.0	81.7	80.6
33	GUE	tf_1	82.5	83.6	81.9
34	GUE	tf_2	75.5	77.3	72.8
35	GUE	tf_3	70.9	69.8	66.9
36	GUE	tf_4	82.4	82.9	81.3
37	snATAC-seq	Human/Mouse brain	30.3	29.5	31.0
38	Multi-SCREEN	Human cCRE	30.1	30.3	29.7
39	NT	H2AFZ	73.7	71.7	72.0
40	NT	H3K27ac	72.7	71.6	70.8
41	NT	H3K27me3	80.0	80.6	79.3
42	NT	H3K36me3	79.2	78.3	78.5
43	NT	H3K4me1	74.1	73.8	72.7
44	NT	H3K4me2	78.4	78.3	76.9
45	NT	H3K4me3	83.3	83.2	82.6
46	NT	H3K9ac	76.4	76.5	74.3
47	NT	H3K9me3	70.3	70.7	70.7
48	NT	H4K20me1	81.3	81.6	81.1
49	NT	enhancers	73.9	73.9	73.0
50	NT	enhancers_types	50.8	50.8	49.2
51	NT	promoter_all	86.1	85.6	85.0
52	NT	promoter_no_tata	87.1	87.0	86.7
53	NT	promoter_tata	84.3	82.0	84.1
54	NT	splice_sites_acceptors	83.7	81.5	81.0
55	NT	splice_sites_all	76.4	72.8	70.2
56	NT	splice_sites_donors	83.4	80.6	78.4

Table 20: Per-task F1 (%) for ablation variants.

#	Benchmark	Task	DNAbert2	NT	Grover
1	GBM	demo_coding_vs_intergenomic_seqs	90.3	89.0	91.5
2	GBM	demo_human_or_worm	95.8	93.8	96.4
3	GBM	drosophila_enhancers_stark	69.1	68.9	69.7
4	GBM	dummy_mouse_enhancers_ensembl	74.8	75.1	73.9
5	GBM	human_enhancers_cohn	73.1	71.8	73.8
6	GBM	human_enhancers_ensembl	89.4	83.5	90.6
7	GBM	human_ensembl_regulatory	90.6	91.5	90.8
8	GBM	human_nontata_promoters	92.9	85.5	95.1
9	GBM	human_ocr_ensembl	73.8	72.1	77.7
10	GUE	H3	85.6	83.0	87.5
11	GUE	H3K14ac	69.0	65.6	70.8
12	GUE	H3K36me3	71.3	68.1	72.2
13	GUE	H3K4me1	68.9	67.4	69.2
14	GUE	H3K4me2	66.5	65.8	66.4
15	GUE	H3K4me3	63.1	58.9	64.9
16	GUE	H3K79me3	79.1	76.2	79.5
17	GUE	H3K9ac	74.5	72.0	75.7
18	GUE	H4	87.9	83.6	87.2
19	GUE	H4ac	67.0	63.9	69.9
20	GUE	mouse_0	66.4	65.4	71.9
21	GUE	mouse_1	89.5	86.1	90.2
22	GUE	mouse_2	87.5	81.2	86.4
23	GUE	mouse_3	75.1	72.5	83.1
24	GUE	mouse_4	65.6	60.6	70.6
25	GUE	prom_300_all	90.5	92.0	91.3
26	GUE	prom_300_notata	95.1	94.6	94.9
27	GUE	prom_300_tata	78.6	82.0	78.3
28	GUE	prom_core_all	80.5	80.3	80.7
29	GUE	prom_core_notata	82.3	81.3	82.8
30	GUE	prom_core_tata	75.9	83.7	79.9
31	GUE	splice_reconstructed	82.9	83.1	87.2
32	GUE	tf_0	82.5	81.0	83.9
33	GUE	tf_1	84.8	82.7	84.4
34	GUE	tf_2	78.3	73.2	80.8
35	GUE	tf_3	69.5	67.6	73.1
36	GUE	tf_4	84.1	81.6	85.8
37	snATAC-seq	Human/Mouse brain	33.2	32.8	38.1
38	Multi-SCREEN	Human cCRE	30.7	30.3	32.6
39	NT	H2AFZ	73.5	74.4	73.7
40	NT	H3K27ac	73.1	71.3	73.6
41	NT	H3K27me3	80.0	80.6	80.7
42	NT	H3K36me3	80.3	78.9	80.1
43	NT	H3K4me1	73.9	74.8	74.5
44	NT	H3K4me2	78.5	78.4	78.5
45	NT	H3K4me3	83.3	82.8	84.1
46	NT	H3K9ac	76.7	74.1	77.2
47	NT	H3K9me3	72.1	68.5	71.3
48	NT	H4K20me1	81.4	80.5	81.5
49	NT	enhancers	73.8	75.1	75.1
50	NT	enhancers_types	61.3	47.5	81.2
51	NT	promoter_all	86.7	85.9	87.0
52	NT	promoter_no_tata	86.8	87.7	87.4
53	NT	promoter_tata	85.8	86.5	86.0
54	NT	splice_sites_acceptors	85.6	96.6	87.5
55	NT	splice_sites_all	79.7	95.7	87.2
56	NT	splice_sites_donors	84.0	97.5	91.5

Table 21: Per-task Precision (%) for tokenizer-strategy comparison.

#	Benchmark	Task	w/o Len	w/o Part	w/o Pri
1	GBM	demo_coding_vs_intergenomic_seqs	90.4	90.6	90.1
2	GBM	demo_human_or_worm	95.3	95.8	95.6
3	GBM	drosophila_enhancers_stark	76.1	74.6	70.9
4	GBM	dummy_mouse_enhancers_ensembl	73.1	73.7	79.5
5	GBM	human_enhancers_cohn	74.4	73.3	73.7
6	GBM	human_enhancers_ensembl	88.5	88.8	87.7
7	GBM	human_ensembl_regulatory	90.3	89.7	89.6
8	GBM	human_nontata_promoters	92.7	93.5	92.9
9	GBM	human_ocr_ensembl	74.0	74.7	73.6
10	GUE	H3	85.0	85.1	83.3
11	GUE	H3K14ac	69.6	69.3	67.2
12	GUE	H3K36me3	71.1	71.3	70.4
13	GUE	H3K4me1	68.7	69.5	68.3
14	GUE	H3K4me2	66.9	66.3	65.7
15	GUE	H3K4me3	62.9	62.2	63.0
16	GUE	H3K79me3	78.7	79.9	77.6
17	GUE	H3K9ac	73.2	73.5	72.2
18	GUE	H4	86.6	86.3	85.6
19	GUE	H4ac	67.1	67.2	66.8
20	GUE	mouse_0	66.7	68.2	64.3
21	GUE	mouse_1	88.6	88.8	88.0
22	GUE	mouse_2	82.0	82.9	84.3
23	GUE	mouse_3	72.1	84.2	77.0
24	GUE	mouse_4	66.4	65.6	65.3
25	GUE	prom_300_all	91.2	91.2	90.7
26	GUE	prom_300_notata	94.5	95.4	94.9
27	GUE	prom_300_tata	79.5	76.0	80.3
28	GUE	prom_core_all	81.0	80.9	80.0
29	GUE	prom_core_notata	82.4	82.7	82.3
30	GUE	prom_core_tata	78.6	81.3	76.5
31	GUE	splice_reconstructed	82.3	80.8	80.9
32	GUE	tf_0	83.4	82.2	81.3
33	GUE	tf_1	83.6	84.4	82.9
34	GUE	tf_2	77.1	77.3	73.6
35	GUE	tf_3	71.2	72.0	68.8
36	GUE	tf_4	82.4	83.0	81.3
37	snATAC-seq	Human/Mouse brain	35.4	35.9	34.4
38	Multi-SCREEN	Human cCRE	31.1	30.9	30.3
39	NT	H2AFZ	73.9	73.5	72.2
40	NT	H3K27ac	73.4	72.1	71.4
41	NT	H3K27me3	80.2	80.7	80.7
42	NT	H3K36me3	79.5	78.3	78.6
43	NT	H3K4me1	74.4	74.5	73.8
44	NT	H3K4me2	78.6	78.7	77.6
45	NT	H3K4me3	83.3	83.2	82.6
46	NT	H3K9ac	76.5	76.6	74.6
47	NT	H3K9me3	70.4	70.9	70.7
48	NT	H4K20me1	81.3	81.7	81.1
49	NT	enhancers	74.1	73.9	73.2
50	NT	enhancers_types	70.5	70.3	69.7
51	NT	promoter_all	86.7	85.6	85.0
52	NT	promoter_no_tata	87.1	87.1	87.0
53	NT	promoter_tata	84.3	82.0	84.1
54	NT	splice_sites_acceptors	83.7	81.6	81.0
55	NT	splice_sites_all	76.5	72.8	70.8
56	NT	splice_sites_donors	83.5	80.6	78.7

Table 22: Per-task Precision (%) for ablation variants.

#	Benchmark	Task	DNAbert2	NT	Grover
1	GBM	demo_coding_vs_intergenomic_seqs	90.2	89.0	91.5
2	GBM	demo_human_or_worm	95.8	93.8	96.4
3	GBM	drosophila_enhancers_stark	68.5	69.0	68.7
4	GBM	dummy_mouse_enhancers_ensembl	73.8	75.4	73.9
5	GBM	human_enhancers_cohn	72.8	71.7	73.6
6	GBM	human_enhancers_ensembl	89.3	83.3	90.5
7	GBM	human_ensembl_regulatory	89.8	90.7	90.3
8	GBM	human_nontata_promoters	93.3	85.6	95.5
9	GBM	human_ocr_ensembl	73.8	72.0	77.7
10	GUE	H3	85.5	82.9	87.4
11	GUE	H3K14ac	68.7	64.8	70.5
12	GUE	H3K36me3	71.4	68.2	72.4
13	GUE	H3K4me1	68.7	66.7	68.9
14	GUE	H3K4me2	63.8	63.8	63.9
15	GUE	H3K4me3	63.1	58.1	65.0
16	GUE	H3K79me3	78.9	75.8	79.5
17	GUE	H3K9ac	74.8	72.2	75.7
18	GUE	H4	87.8	83.3	85.8
19	GUE	H4ac	67.0	63.4	69.8
20	GUE	mouse_0	66.4	65.3	71.9
21	GUE	mouse_1	89.4	85.9	90.2
22	GUE	mouse_2	87.5	81.1	86.3
23	GUE	mouse_3	74.9	72.4	82.8
24	GUE	mouse_4	65.6	60.6	70.5
25	GUE	prom_300_all	90.5	92.0	91.2
26	GUE	prom_300_notata	95.1	94.6	94.9
27	GUE	prom_300_tata	78.6	82.1	78.3
28	GUE	prom_core_all	80.5	80.2	80.7
29	GUE	prom_core_notata	82.2	81.2	82.8
30	GUE	prom_core_tata	75.9	83.7	79.9
31	GUE	splice_reconstructed	84.0	85.9	87.6
32	GUE	tf_0	81.7	80.3	83.2
33	GUE	tf_1	84.5	81.9	83.7
34	GUE	tf_2	77.5	72.7	79.0
35	GUE	tf_3	68.2	67.1	72.2
36	GUE	tf_4	84.0	81.4	85.8
37	snATAC-seq	Human/Mouse brain	30.4	30.9	33.7
38	Multi-SCREEN	Human cCRE	31.6	31.0	33.3
39	NT	H2AFZ	73.5	73.5	73.4
40	NT	H3K27ac	73.1	71.0	73.3
41	NT	H3K27me3	80.0	80.0	80.4
42	NT	H3K36me3	80.1	78.5	80.1
43	NT	H3K4me1	73.7	73.0	73.9
44	NT	H3K4me2	78.2	77.4	78.2
45	NT	H3K4me3	83.1	82.8	83.6
46	NT	H3K9ac	76.7	73.6	77.2
47	NT	H3K9me3	71.9	67.9	71.3
48	NT	H4K20me1	81.4	80.3	81.4
49	NT	enhancers	73.8	74.5	75.0
50	NT	enhancers_types	50.7	48.9	50.0
51	NT	promoter_all	86.7	85.9	86.9
52	NT	promoter_no_tata	86.8	87.7	87.3
53	NT	promoter_tata	85.8	86.4	86.0
54	NT	splice_sites_acceptors	85.6	96.6	87.5
55	NT	splice_sites_all	79.6	95.7	87.2
56	NT	splice_sites_donors	84.0	97.5	91.4

Table 23: Per-task Recall (%) for tokenizer-strategy comparison.

#	Benchmark	Task	w/o Len	w/o Part	w/o Pri
1	GBM	demo_coding_vs_intergenomic_seqs	90.4	90.6	90.1
2	GBM	demo_human_or_worm	95.2	95.8	95.6
3	GBM	drosophila_enhancers_stark	76.2	74.6	71.0
4	GBM	dummy_mouse_enhancers_ensembl	73.1	73.3	79.6
5	GBM	human_enhancers_cohn	74.3	73.2	73.6
6	GBM	human_enhancers_ensembl	88.4	88.6	87.6
7	GBM	human_ensembl_regulatory	89.4	88.8	88.8
8	GBM	human_nontata_promoters	93.1	93.9	93.2
9	GBM	human_ocr_ensembl	73.9	74.6	73.5
10	GUE	H3	84.9	84.9	83.3
11	GUE	H3K14ac	69.1	69.7	67.5
12	GUE	H3K36me3	71.2	71.5	70.3
13	GUE	H3K4me1	68.3	69.1	68.0
14	GUE	H3K4me2	64.1	63.1	62.7
15	GUE	H3K4me3	62.5	61.6	62.7
16	GUE	H3K79me3	78.7	79.9	77.7
17	GUE	H3K9ac	73.2	73.7	72.5
18	GUE	H4	86.3	86.3	85.3
19	GUE	H4ac	67.0	66.8	66.8
20	GUE	mouse_0	66.7	68.0	64.3
21	GUE	mouse_1	88.6	88.8	88.0
22	GUE	mouse_2	82.0	82.9	84.1
23	GUE	mouse_3	72.0	84.1	77.0
24	GUE	mouse_4	66.1	65.5	65.1
25	GUE	prom_300_all	91.2	91.2	90.7
26	GUE	prom_300_notata	94.5	95.4	94.9
27	GUE	prom_300_tata	79.3	75.8	80.2
28	GUE	prom_core_all	80.9	80.9	80.0
29	GUE	prom_core_notata	82.4	82.7	82.3
30	GUE	prom_core_tata	77.9	80.8	76.5
31	GUE	splice_reconstructed	82.1	80.4	80.3
32	GUE	tf_0	83.0	81.8	80.7
33	GUE	tf_1	82.6	83.7	82.0
34	GUE	tf_2	75.8	77.3	73.0
35	GUE	tf_3	71.0	70.4	67.5
36	GUE	tf_4	82.4	82.9	81.3
37	snATAC-seq	Human/Mouse brain	33.0	32.9	32.5
38	Multi-SCREEN	Human cCRE	31.2	31.3	30.9
39	NT	H2AFZ	73.8	72.2	72.1
40	NT	H3K27ac	72.8	71.7	71.0
41	NT	H3K27me3	80.0	80.6	79.6
42	NT	H3K36me3	79.3	78.3	78.5
43	NT	H3K4me1	74.2	74.0	73.0
44	NT	H3K4me2	78.4	78.4	76.9
45	NT	H3K4me3	83.3	83.2	82.6
46	NT	H3K9ac	76.4	76.5	74.3
47	NT	H3K9me3	70.3	70.7	70.7
48	NT	H4K20me1	81.3	81.6	81.1
49	NT	enhancers	73.9	73.9	73.0
50	NT	enhancers_types	50.9	50.8	49.8
51	NT	promoter_all	86.3	85.6	85.0
52	NT	promoter_no_tata	87.1	87.1	86.7
53	NT	promoter_tata	84.3	82.0	84.1
54	NT	splice_sites_acceptors	83.7	81.5	81.0
55	NT	splice_sites_all	76.4	72.8	70.2
56	NT	splice_sites_donors	83.4	80.6	78.4

Table 24: Per-task Recall (%) for ablation variants.

#	Benchmark	Task	Base	EvoLen	Δ
1	GBM	demo_coding_vs_intergenomic_seqs	82.1	81.8	-0.2
2	GBM	demo_human_or_worm	92.3	92.7	+0.4
3	GBM	drosophila_enhancers_stark	54.3	56.4	+2.1
4	GBM	dummy_mouse_enhancers_ensembl	53.8	64.2	+10.4
5	GBM	human_enhancers_cohn	45.3	50.1	+4.8
6	GBM	human_enhancers_ensembl	79.4	79.3	-0.1
7	GBM	human_ensembl_regulatory	84.2	84.2	+0.0
8	GBM	human_nontata_promoters	89.3	89.9	+0.6
9	GBM	human_ocr_ensembl	51.1	50.6	-0.5
10	GUE	H3	70.7	72.4	+1.6
11	GUE	H3K14ac	39.9	41.8	+1.9
12	GUE	H3K36me3	43.7	45.1	+1.4
13	GUE	H3K4me1	38.2	37.8	-0.4
14	GUE	H3K4me2	29.2	30.6	+1.5
15	GUE	H3K4me3	28.8	28.7	-0.1
16	GUE	H3K79me3	57.1	59.9	+2.7
17	GUE	H3K9ac	47.6	50.4	+2.7
18	GUE	H4	74.4	75.6	+1.1
19	GUE	H4ac	39.4	36.7	-2.7
20	GUE	mouse_0	33.9	45.7	+11.8
21	GUE	mouse_1	79.3	79.4	+0.1
22	GUE	mouse_2	72.6	73.8	+1.2
23	GUE	mouse_3	54.8	59.5	+4.7
24	GUE	mouse_4	32.1	35.9	+3.8
25	GUE	prom_300_all	83.5	83.0	-0.5
26	GUE	prom_300_notata	89.8	90.8	+1.0
27	GUE	prom_300_tata	59.5	60.5	+1.0
28	GUE	prom_core_all	63.6	63.1	-0.5
29	GUE	prom_core_notata	64.5	66.0	+1.5
30	GUE	prom_core_tata	66.4	64.1	-2.3
31	GUE	splice_reconstructed	75.5	75.6	+0.1
32	GUE	tf_0	65.1	65.8	+0.7
33	GUE	tf_1	67.7	68.3	+0.6
34	GUE	tf_2	53.1	57.0	+3.9
35	GUE	tf_3	45.1	46.1	+1.0
36	GUE	tf_4	62.1	68.8	+6.7
37	snATAC-seq	Human/Mouse brain	10.3	13.7	+3.4
38	Multi-SCREEN	Human cCRE	22.7	22.5	-0.2
39	NT	H2AFZ	47.9	49.1	+1.2
40	NT	H3K27ac	46.5	46.0	-0.5
41	NT	H3K27me3	57.5	58.8	+1.3
42	NT	H3K36me3	60.5	61.7	+1.3
43	NT	H3K4me1	45.6	47.9	+2.3
44	NT	H3K4me2	54.3	58.5	+4.2
45	NT	H3K4me3	64.0	66.1	+2.2
46	NT	H3K9ac	52.2	54.6	+2.4
47	NT	H3K9me3	39.4	42.0	+2.6
48	NT	H4K20me1	66.9	69.4	+2.5
49	NT	enhancers	51.1	53.5	+2.4
50	NT	enhancers_types	48.3	46.5	-1.8
51	NT	promoter_all	73.1	74.9	+1.8
52	NT	promoter_no_tata	76.5	77.0	+0.5
53	NT	promoter_tata	75.0	76.9	+1.9
54	NT	splice_sites_acceptors	63.9	68.7	+4.8
55	NT	splice_sites_all	69.2	68.3	-0.8
56	NT	splice_sites_donors	73.9	68.1	-5.8

Table 25: Per-task MCC (%) for the compute-matched comparison at 200k pretraining steps. Both models use identical fine-tuning hyperparameters per task; the only difference is the tokenizer. EvoLen improves on 43 of 56 tasks.

A.7 Stratification Window Size

The 100 bp stratification window balances two competing pressures: shorter windows track base-level phyloP more faithfully but fragment motifs, while longer windows smooth away the conservation signal the partition is meant to capture. Table 26 reports chromosome 1 diagnostics across window sizes. *SD* is the standard deviation of the window-mean phyloP; *Lag-1* is the autocorrelation between adjacent windows; *Agreement* and κ compare base-level conservation labels against the 100 bp reference partition. Dispersion falls monotonically with window size, lag-1 autocorrelation peaks near 100 bp, and label stability is highest at 50–200 bp. The lower panel reports the probability that a motif of a given length is split

across a bin boundary: at 100 bp, 89–95% of 6–12 bp motifs fall entirely within a single bin, whereas motif-scale windows split most of them.

Table 26: PhyloP window-size diagnostics on chromosome 1 (upper) and motif boundary-crossing probability by bin size (lower).

Window (bp)	Windows	SD	Lag-1	Agreement%	κ
5	49,791,284	0.360	0.256	89.12	0.293
10	24,895,642	0.285	0.363	90.74	0.399
25	9,958,256	0.224	0.521	93.45	0.562
50	4,979,128	0.195	0.608	95.67	0.703
100	2,489,564	0.175	0.614	—	—
200	1,244,782	0.157	0.569	95.70	0.699
500	497,912	0.134	0.511	93.18	0.525

Bin (bp)	6 bp motif	8 bp motif	12 bp motif
10	0.500	0.700	1.000
25	0.200	0.280	0.440
50	0.100	0.140	0.220
100	0.050	0.070	0.110
200	0.025	0.035	0.055
500	0.010	0.014	0.022

A.8 Category-Specific Vocabulary Distinctness

The three category-specific BPE vocabularies are not redundant. Table 27 reports pairwise Jaccard overlap and per-pool compression statistics. bp/Tok is the mean number of base pairs per token; H is the unigram Shannon entropy in bits; and H_{norm} is $H / \log_2 |V|$. Conserved and Accelerated vocabularies share only 32–34% of their tokens at every vocabulary size, while Conserved and Neutral overlap substantially more (52–58%), consistent with conservation forming a gradient rather than discrete classes. Normalised entropy is consistently highest for the Neutral pool and lowest for Accelerated, a gap of roughly 0.028 that is stable across vocabulary sizes, indicating that the Accelerated pool concentrates usage on fewer tokens.

Table 27: Pairwise vocabulary Jaccard overlap (upper) and per-pool compression and entropy (lower) for the three category-specific BPEs.

Vocab	Con-Neu	Acc-Con	Acc-Neu
2048	0.579	0.324	0.389
3072	0.523	0.325	0.359
4096	0.547	0.330	0.364
5120	0.564	0.336	0.375

Vocab	Pool	bp/Tok	H	H_{norm}
2048	Conserved	4.747	9.484	0.863
2048	Neutral	4.829	9.541	0.868
2048	Accelerated	4.677	9.254	0.842
3072	Conserved	4.987	9.947	0.859
3072	Neutral	5.083	10.002	0.864
3072	Accelerated	4.912	9.686	0.836
4096	Conserved	5.152	10.262	0.855
4096	Neutral	5.265	10.332	0.861
4096	Accelerated	5.081	9.997	0.833
5120	Conserved	5.279	10.503	0.853
5120	Neutral	5.408	10.586	0.859
5120	Accelerated	5.213	10.240	0.831